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Jurassic location of the Yukon-Tanana terrane from palaeomagnetism of the folded Mississippian Tatlmain batholith and Ragged stock

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SUMMARY

The extensive Yukon-Tanana terrane (YTT) of the northern Canadian Cordillera has been considered controversially to be part of the allochthonous 'Baja BC' microcontinent or of the para-autochthonous North American cratonic margin during the Mesozoic. Palaeomagnetic methods have isolated a very-stable Early Jurassic thermochemical remanent remagnetization in the terrane's felsic Tatlmain batholith and mafic Ragged stock after correction for: (1) rotation from northeast-plunging anticlinal deformation, (2) northerly dipping tectonic tilt of the host rocks and (3) northwestward regional translation on the adjacent Tintina transcurrent fault zone. The resulting 196 ± 6 Ma Tatlmain and Ragged palaeopoles are 64.9°N, 44.8°E (A_{95} = 5.9°) and 64.2°N, 58.5°E (A_{95} = 7.7°), respectively. The YTT palaeopoles support para-autochthonous tectonic models that have the YTT: (1) accreting to North America by the Early Jurassic, (2) undergoing non-significant orogen-perpendicular shortening by mid-Early Cretaceous from thrust-faulting, and, then (3) undergoing significant orogen-parallel northward translation of ∼500 km from mid-Early Cretaceous to the Eocene. In contrast, the palaeopoles for Stikinia and Quesnellia of the Intermontane Belt show progressive northwestward translation relative to North America by ∼1000 km and a rotation of ∼55° cw since mid-Early Cretaceous. We speculate that ∼500 km of the northward translation is related to dextral motion on the Tintina and Northern Rocky Mountain Trench fault in British Columbia, and that the clockwise rotation is related to upper crustal tectonics in both Yukon and southern British Columbia.

Key words: North America; Palaeomagnetism; Remagnetization; Continental tectonics: strike-slip and transform; Folds and folding.

1 INTRODUCTION

The Yukon-Tanana terrane (YTT) of the North American Cordillera crops out over large areas of east-central Alaska, southwestern Yukon and northernmost British Columbia (Fig. 1; Colpron *et al*. 2006a). As the result of the YTT's complex deformational and metamorphic history, understanding the geological evolution of these mid- to late Palaeozoic rocks has long proven challenging (Tempelman-Kluit 1976; Colpron *et al.* 2006a). Thus hypotheses for the original location of the YTT's continental margin, arc and marginal basin assemblages relative to North America range from autochthonous (e.g. Hansen 1990) or para-autochthonous (e.g. Hansen & Dusel-Bacon 1998) to allochthonous (e.g. Mortensen 1992). Early palaeomagnetic work on late Palaeozoic strata in the Slide Mountain terrane and Late Cretaceous strata in the YTT and overlapping Intermontane Belt (IMB) terranes was interpreted to show that the YTT was an exotic allochthonous terrane that was translated some 1500–3500 km northwards relative to North America before accreting to the continent (e.g. Richards *et al.* 1993; Johnston 2001; Enkin *et al.* 2006). Conversely, geologic evidence and palaeomagnetic work on several Cretaceous plutons have indicated that the YTT is pericratonic, having travelled ∼500 km northwards relative to North America, mostly as a result of Palaeocene–Eocene dextral transcurrent fault displacement on the inboard Tintina fault zone (Gabrielse 1985; Symons *et al.* 2005; 2015, 2017; Gabrielse *et al.* 2006; McCausland *et al.* 2006; Symons & Kawasaki 2011). Heretofore, the three previously published Jurassic YTT palaeopoles from the Big Creek, Granite Mountain and Taylor Mountain plutons have been too scattered to reliably locate the YTT during the latest Triassic to mid-Cretaceous interval (Symons *et al*. 2000; McCausland *et al.* 2001, 2009).

When collected in the summer of 2000, the Ragged gabbro-syenite stock was thought to be Jurassic and to intrude the late

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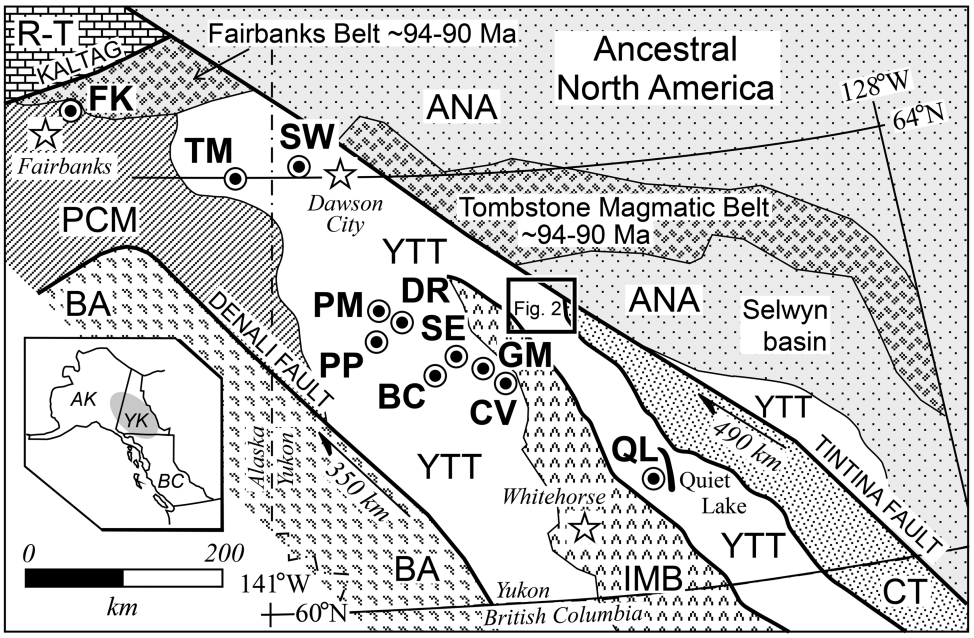


Figure 1. Location of the Yukon-Tanana terrane (YTT) and of reported palaeomagnetic studies in the YTT. More detailed information for the Ragged stock and Tatlmain batholith study area is illustrated in Figs 2 and 3. ANA, Ancestral North America; BA, Baja Alaska (northern Insular Belt); BC, Big Creek batholith; CT, Cassiar terrane; CV, Carmacks volcanics (mean location of scattered collection sites); DR, Dawson Range batholith; FK, Fort Knox stock; GM, Granite Mountain batholith; IMB, Intermontane Belt terranes; PCM, para-autochthonous YTT continental margin; PM, Prospector Mountain stock; PP, Pattison pluton; QL, Quiet Lake batholith; R-T, Ruby and Tozitna terranes; SE, Seymour Creek stock; SW, Swede Dome stock; TM, Taylor Mountain batholith. References for the collections are given in Table 2. Detailed digital geologic maps for Yukon are available from Colpron *et al.* (2016a,b)

Palaeozoic Tatlmain quartz-diorite batholith. These two units were chosen for palaeomagnetic study because of their unfoliated massive texture, their minimal evidence of thermal alternation and the promise that U/Pb zircon age dates in progress would provide reliable ages for the stock and batholith. However, shortly after the collected specimens were measured palaeomagnetically, Ragged stock was found to yield a U/Pb zircon date of 356 ± 1 Ma, to be emplaced into the adjacent Snowcap Assemblage and to be older than the ∼339 ± 1 Ma Tatlmain batholith (Colpron *et al*. 2002, 2006a). Since 2002, the apparent polar wander path (APWP) for the YTT has been well defined back to the mid-Early Cretaceous by several palaeopoles that show the YTT to be pericratonic since then (Symons *et al*. 2005, 2017). In this palaeomagnetic study we interpret the Ragged and Tatlmain intrusions to carry an Early Jurassic overprint remanence. When the two overprint palaeopoles are added to the three previously published latest Triassic to Early Jurassic palaeopoles, they show the YTT to have been accreted to the North American plate by the mid-Early Jurassic, which provides a constraint on tectonic models for the post-Triassic evolution of the YTT.

2 GEOLOGY

Ragged stock and Tatlmain batholith intrude the YTT in the Glenlyon map area (NTS 105 L/1–7, 11–14; 115I/9, 16) of central Yukon (Fig. 1). Early geological mapping in the area was done by Campbell (1967), Templeman-Kluit (1984), Oliver (1996) and Oliver & Mortensen (1998). The following geological summary of the sampled study area is taken mostly from extensive mapping and geochemical research reported by Colpron et al. (2002, 2005, 2006a,b, 2007a,b, 2016a,b).

2.1 Geology of the study area

In the northern part of the Glenlyon map area, the YTT study area crops out as a ∼50-km-wide northwest-trending tectonic block (Fig. 2). The block is separated on its northeast side from Cambro-Ordovician phyllites in the Selwyn basin of ancestral North America by the dextral transcurrent Tintina fault zone. On its eastern side, the dextral transcurrent Big Salmon fault separates the block from the southeastward continuation of the YTT, a thin slice of the Slide Mountain terrane and the Cassiar terrane. The pericratonic Cassiar terrane is represented by Silurian to Mississippian dolomites of the Askin Group and black shales of the Earn Group. On its southwestern side, the tectonic block is separated from the Stikine terrane of the Intermontane Belt (IMB) by the Tadru transcurrent fault and linked Needlerock northwest-verging thrust fault. The adjacent Stikine terrane contains Early Jurassic basalts of the Semenof Formation and the mid-Jurassic McGregor and Tachun granodiorite batholiths.

In the studied tectonic block, psammitic schist of the basal Devonian and older (?) Snowcap Assemblage forms the basement of the YTT (Colpron *et al.* 2006a; Morneau *et al.* 2017). The Early Mississippian Ragged gabbro-syenite stock intrudes the Snowcap Assemblage, and both units are unconformably overlain by the Early to Middle Mississippian Finlayson Assemblage (Gradstein *et al.* 2004). This assemblage contains arkosic grit and psammite of the basal Drury Formation, quartzite of the overlying Pelmac Formation, and volcanic rocks of the upper Little Kalzas Formation. Both the Snowcap and Finlayson assemblages have been intruded by granodiorite and tonalite of the Early to Middle Mississippian Little Kalzas plutonic suite and by granite and granodiorite of the Middle Mississippian Tatlmain plutonic suite.

Ragged stock has a surface area of ∼50 km^2 (Fig. 3). It is composed of unfoliated coarse-grained augite gabbro and K-feldspar

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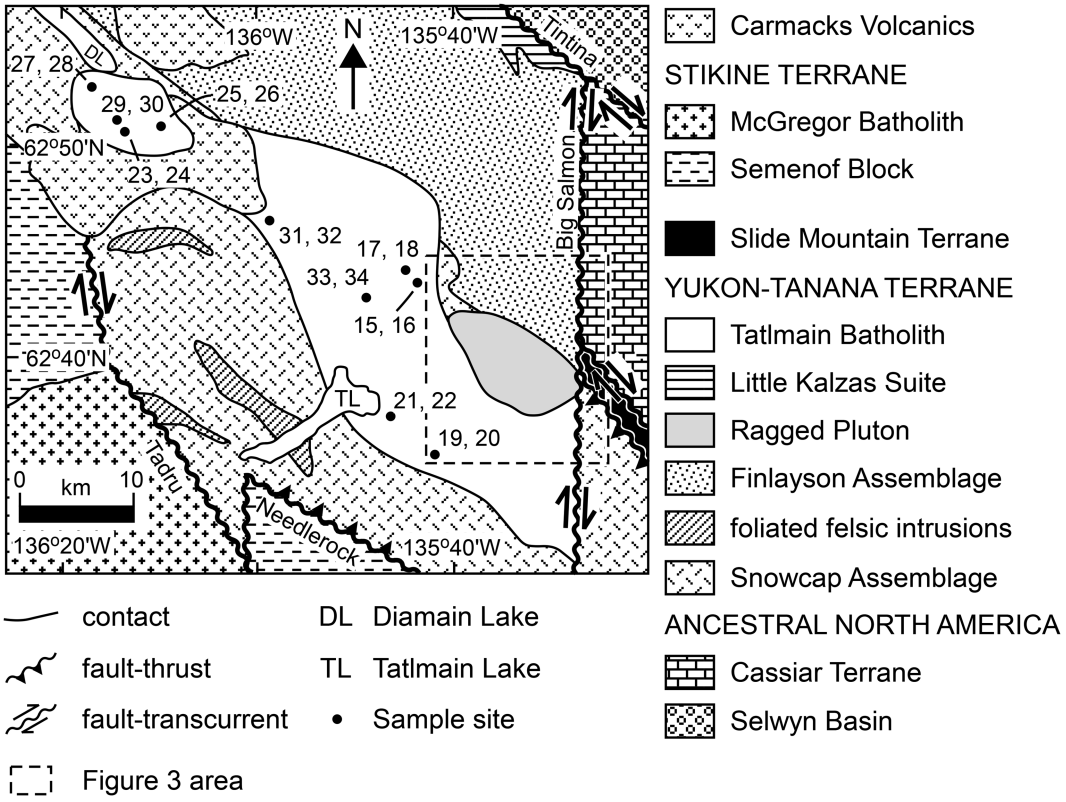


Figure 2. Geology and sampling site locations for the Tatlmain batholith.

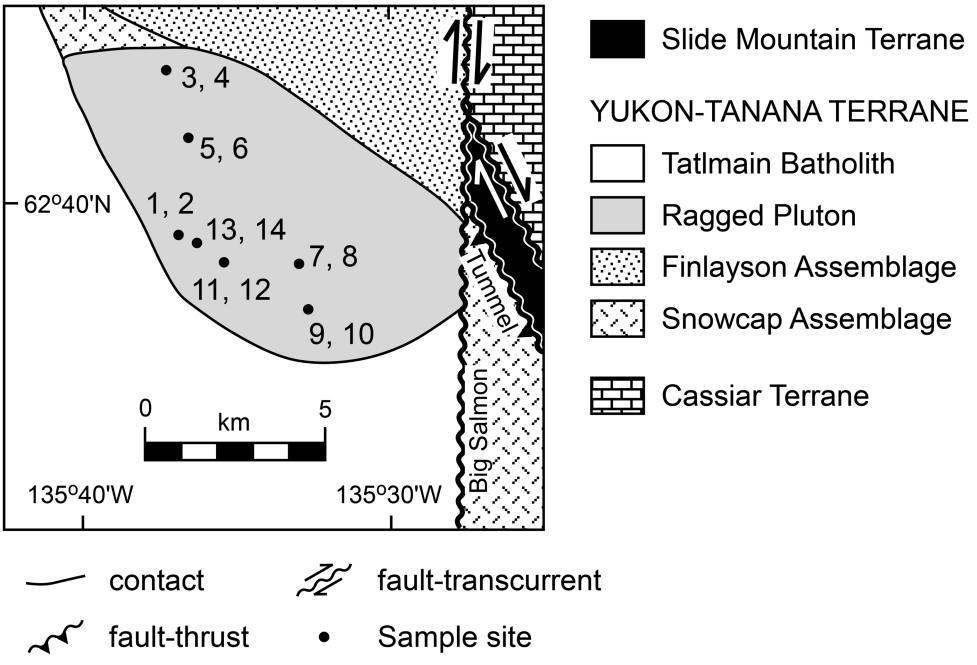


Figure 3. Geology and sampling site locations for Ragged stock.

porphyritic monzonite (Colpron *et al.* 2006a). The stock is inferred to intrude the Snowcap Assemblage and to be intruded by granite of the adjacent Tatlmain batholith. Two concordant zircon fractions from the stock have given $^{206}Pb/^{238}U$ zircon ages of 356.4 ± 1.0 Ma that are confirmed by a $^{206}Pb/^{238}U$ titanite date of 357.9 ± 1.9 Ma (Colpron *et al.* 2006a). The stock's trace-element geochemical characteristics are typical of volcanic arc granitoids.

The Tatlmain batholith covers an outcrop area of ~500 km^2 (Fig. 2). The batholith is composed of unfoliated homogenous coarse-grained biotite (±hornblende) quartz diorite to granite. The

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batholith intrudes deformed rocks of the Snowcap Assemblage, Ragged stock and the Pelmac quartzite. The batholith has given $^{206}Pb/^{238}U$ zircon ages of 339.5 ± 1.3, 338.5 ± 1.0 and 339.4 ± 0.9 Ma as well as a $^{40}Ar/^{39}Ar$ hornblende plateau age of 343.7 ± 3.2 Ma (Colpron *et al.* 2006a). The batholith has also imposed a contact metamorphic aureole on both the Pelmac quartzite and Snowcap Assemblage. Lithoprobe seismic data have been interpreted to show that batholiths and stocks in the YTT and adjacent terranes are typically 'pancake-shaped slabs' that are broadly conformable with the host strata and a few kilometres thick (Cook *et al.* 2004). The Tatlmain plutonic suite is deemed to be the subvolcanic root of the overlying volcanics of the Little Salmon Formation of the uppermost Finlayson Assemblage that crop out to the southeast of the study area and that have given a $^{206}Pb/^{238}U$ zircon age of 340.2 ± 2.1 Ma (Colpron 2001; Colpron *et al.* 2006a).

Gently dipping (∼5°NE) mafic flows of the Late Cretaceous (∼70 Ma) Carmacks volcanics outcrop over an area of ∼150 km^2 in the study area around the northwest end of the Tatlmain batholith (Fig. 2). The volcanics unconformably overlie both the batholith and adjacent Pelmac Formation.

2.2 Metamorphism in the study area

The Snowcap Assemblage typically contains metamorphic rocks of the amphibolite facies that experienced Devonian or older (?) metamorphism (Colpron *et al.* 2006a). In contrast, the rocks of the overlying Finlayson Assemblage have experienced greenschist metamorphism only, excluding contact metamorphism associated with emplacement of the Little Kalzas plutonic suite and Tatlmain batholith (Colpron *et al.* 2006a; Staples *et al.* 2016). The Tatlmain batholith has also given a $^{40}Ar/^{39}Ar$ hornblende plateau age of 343.7 ± 3.2 Ma that indicates rapid cooling to below the *ca.* 530 °C closure temperature of Ar in hornblende (McElhinny & McFadden 2000). However, nine $^{40}Ar/^{39}Ar$ ages and one K/Ar age with a mean age of 196 ± 6 Ma have been reported for muscovite and biotite from the Tatlmain batholith and Finlayson Assemblage in the study area (Breitsprecher *et al.* 2004; Colpron *et al.* 2007b). This cluster of ages indicates greenschist metamorphism, uplift, erosion and cooling of the tectonic block from about 500 to 300 °C between about 205 and 190 Ma. Also in the Early Jurassic, the Stikine terrane of the IMB impacted on the southwest side of the YTT, resulting in regional folding, thrust faulting and the emplacement of the large Tatchun granitic batholith along the southwest side of the studied tectonic block (Fig. 2). Tatchun batholith has yielded seven $^{40}Ar/^{39}Ar$ biotite dates with a mean of 185 ± 5 Ma (Colpron *et al.* 2007b).

2.3 Post-Finlayson assemblages of the YTT

Outside of our study area, the Finlayson Assemblage is unconformably overlain by the Klinkit Assemblage. It includes Middle Mississippian to Early Permian intermediate to mafic volcanoclastic, volcanic and limestone rocks that are deemed to record continental arc magmatism and intra-arc rifting environments (Colpron 2001; Colpron *et al.* 2006a). The adjacent oceanic Slide Mountain Assemblage is composed of Mississippian to Early Permian chert, argillite and basalt, and hence is about coeval with the Klinquit Assemblage. Both the Slide Mountain and Klinkit assemblages are unconformably overlain by the Klondike Assemblage, which consists of Middle to Late Permian calc-alkaline felsic rocks with minor mafic volcanic rocks and comagmatic plutons. This assemblage has the geochemical signature of continental arc magmatism, and it corresponds in time to the Klondike orogeny (∼275 to ∼260 Ma) when the Slide Mountain oceanic terrane subducted under the YTT (Mortensen 1992; Nelson *et al.* 2006; Beranek & Mortensen 2011; Parsons *et al.* 2019). A cover sequence of calcareous siltstones, sandstones and shales unconformably overlaps the Yukon-Tanana, Slide Mountain, Cassiar and Selwyn Basin terranes. Deposition began in latest Permian but occurred predominantly in the Middle and Late Triassic (Colpron *et al.* 2006b). Pressure–temperature–time (*P–T–t*) measurements on garnet and monazite crystals associated with amphibolite-facies metamorphism indicate that ductile deformation of the YTT occurred mostly between ∼264 and ∼254 Ma in the Klondike orogeny at pressures of ∼9 kbar and temperatures of ∼600 °C during plutonism (Berman *et al.* 2007; Staples *et al.* 2016).

2.4 Post-Middle Jurassic deformation

About 70 km south of our study area, Lithoprobe ran reflection seismic and gravity profiles across the Whitehorse Trough (Calvert *et al.* 2017). The profiles display a synform that is > 100 km across, ∼18–20 km deep and filled with deformed Mesozoic strata of the Intermontane Belt (IMB). The synform overlies about 16–18 km of YTT rocks in large-scale imbricate structures and the Cassiar terrane of the North American craton. In the synform are folded formations of the Triassic Lewes River Group, Lower to Middle Jurassic Laberge Group and Middle Jurassic to Lower Cretaceous Tantalus Formation, which suggests significant compression after the mid-Cretaceous (Parson *et al.* 2018). Further south in Yukon and northern British Columbia, the Lewes River Group is an overlap assemblage on the Stikinia, Quesnellia and Cache Creek terranes (Colpron *et al.* 2015; Staples *et al.* 2016).

Three major Cretaceous-to-Eocene plate tectonic processes may have been important in the evolution of the YTT. One is the easterly subduction and collision of the Farallon and Kula plates with North America and the related intrusion of the Coast Plutonic Complex (CPC) from ∼140 to ∼40 Ma along the western margin of the YTT (Engebretson *et al.* 1985). The second process is the counterclockwise rotation of the Arctic Alaskan plate to open the Canada basin in the Arctic Ocean, and to close against the western side of Yukon from ∼150 Ma to ∼80 Ma (Plafker & Berg 1994). The third process is the ∼500 to > 1000 km of dextral shear displacement both along and between the Tintina and Coast Shear Zones on either side of the YTT from the mid-Early Cretaceous to the Eocene (Gabrielse et al. 2006; McCausland *et al.* 2006; Symons *et al.* 2006; 2017).

2.5 Plate tectonic models for the YTT

Parsons *et al.* (2019) evaluated and assigned the published tectonic models for the YTT into three categories, i.e. the Klondike Orogeny, Intra-Oceanic Arc and Cordilleran Ribbon models. The Klondike Orogeny model and its variants suggest that the YTT is an allochthonous continental island arc terrane that rifted from the Laurentian margin in the Devonian-Early Mississippian, travelled northwestwards (NW) in the Panthalassa Ocean to deposit the Slide Mountain Ocean on its eastern side, then travelled northeastwards to close the Slide Mountain Ocean, and finally obducted onto, and accreted with, the North American plate in the Late Permian Klondike orogeny with continuing sedimentation and magmatism to the Early Jurassic (Mortensen 1992; Colpron *et al.* 2006b, 2007b, 2015; Nelson *et al.* 2006, 2013; Beranek & Mortensen 2011; Staples *et al.*

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2016). The Intra-Oceanic Arc tectonic model for the YTT hypothesizes that the mid-Permian Dunite Peak ophiolite was part of an island arc that formed at 270–280 Ma in the Slide Mountain ocean, that the YTT subducted eastwards under the arc and accreted to it in the latter Permian from 272 to 252 Ma, and that the arc then collided with, and accreted to, the western margin of Laurentia after the Middle Triassic with continuing orogenesis through to the Middle Jurassic (van Staal *et al.* 2018; Parsons *et al.* 2019).

The Cordilleran Ribbon model proposes that the YTT and IMB were part of SAYBIA, an ∼8000 km long by ≤600 km wide continent that formed along and offshore of western North America in the Panthalassa ocean (Johnston 2001). SAYBIA was assembled by Late Triassic–Early Jurassic accretion of the IMB overthrusting the YTT and was stitched together by ∼186 Ma plutons at a location ∼3000 km to the south and offshore of Mexico (Johnston & Erdmer 1995; Johnston 2008; Hildebrand 2009). The model proposes that SAYBIA was translated slowly northwards from the Early Jurassic to Early Cretaceous, colliding initially with North America at ∼120 Ma. From ∼120 to ∼40 Ma, dextral transpression on strike-slip faults moved the YTT and IMB northwards by ∼2000 km, bulldozed a large region of the continental foreland eastwards on thrust faults and folds to form the Mackenzie and Rocky Mountains, and intruded a series of eastward-younging (140–40 Ma) plutons to create the Coast Ranges.

3 METHODS

3.1 Sampling

The Ragged and Tatlmain intrusions are located in a remote mountainous region of Yukon, requiring helicopter access with few potential landing locations because of recessive weathering. Therefore, two well-separated sites were sampled at each landing location. At each site, six cores were drilled and oriented using a solar compass in most instances or a magnetic compass with topographic checks in a few instances. A total of 34 sites were collected with 14 from the Ragged stock and 20 from the Tatlmain batholith. From 7 to 16 standard paleomagnetic specimens were sliced from the cores for each site for palaeomagnetic measurement, yielding a total of 412 specimens.

3.2 Measurements

After storage for several months in a magnetically shielded room, the natural remanent magnetization (NRM) of each specimen was measured using an automated vertical 2G Enterprises 3-axis DC SQUID cryogenic magnetometer with a sensitivity of $\sim 2 \times 10^{-6}$ A m^{-1} for a standard specimen. Two example specimens per site were subjected to alternating field (AF) demagnetization in 14 steps up to 140 mT using a Sapphire Instruments SI-4 demagnetizer. Next, one of these two specimens per site was subjected to saturation isothermal remanent magnetization (SIRM) tests by pulse magnetizing it in a direct current (DC) field in 13 steps up to 900 mT, which is taken to give a saturation remanence (J_{900}), using a Sapphire Instruments SI-6 DC pulse magnetizer and then AF demagnetizing it in eight steps up to 150 mT. A second pair of example specimens per site were subjected to thermal (TH) demagnetization in 16 steps up to 600 or 610 °C using a Magnetic measurements MMTD-80 oven. Following examination of the data from the AF and TH step demagnetization of the example specimens, up to eight additional specimens per site were TH demagnetized in 12 steps up to 610 °C, except that a small number of glued specimens were AF demagnetized in 10 steps up to 80 mT. The specimen characteristic remanent magnetization (ChRM) directions were obtained using vector density plots (Kamb 1959), orthogonal vector component plots (Zijderveld 1967) and least-squares vector fitting (Kirschvink 1980) methods. The great-circle technique was not used to obtain any ChRM directions (Bailey & Halls 1984). The specimen ChRM directions were then averaged to obtain the site mean ChRM directions and they in turn averaged to get the unit mean ChRM directions following Fisher (1953).

4 RESULTS

4.1 Natural remanent magnetization (NRM)

The 173 measured specimens of mafic syenite from Ragged stock have NRM intensities that range between 2.17×10^{-4} and 5.91×10^{0} A m^{-1} with median (M) and quartile (Q) values of $M = 2.25 \times 10^{-3}$ ($Q_1 = 1.01 \times 10^{-3}$, $Q_3 = 2.02 \times 10^{-2}$) A m^{-1} (Table 1). In contrast, the 239 measured specimens of granodiorite from Tatlmain batholith have NRM intensities that are about an order of magnitude weaker with values that range between 3.36×10^{-5} and 4.02×10^{-1} A m^{-1} with $M = 1.34 \times 10^{-3}$ ($Q_1 = 3.64 \times 10^{-4}$, $Q_3 = 5.67 \times 10^{-3}$) A m^{-1} (Table 1).

4.2 Progressive AF demagnetization

On progressive AF step demagnetization, after removal of the viscous remanent magnetization (VRM) by ∼15 mT, the specimens from both the Ragged and Tatlmain plutons retain a consistent remanence direction up to the 20–50 mT step but have lost ∼75 ± 10 per cent of their NRM intensity (Fig. 4). This behaviour is typically indicative of titanomagnetite, coarse-grained magnetite or possibly pyrrhotite (Dunlop & Özdemir 1997). After this initial intensity decrease, the specimens of the two plutons behave differently. The Ragged stock specimens mostly exhibit erratic changes in both remanence direction and intensity, which is indicative of the presence of either titanomagnetite or coarse-grained magnetite with a significant initial VRM component (Figs 4a and b). This behaviour is well illustrated by Kamb (1959) plots where the remanence vectors for the 15–30 mT steps form a highly coherent population (Fig. 5a) that degrades rapidly by the 40–70 mT steps (Fig. 5b) and becomes entirely scattered by the 80 mT step and above.

After the initial rapid intensity decrease up to 20–50 mT, the Tatlmain specimens mostly continue on further AF step demagnetization to decay fairly smoothly in direction but more slowly in intensity (Figs 4c and d) and to define a coherent population of vector directions up to 160 mT (Figs 5c and d). This pattern of remanence of AF demagnetization suggests that the Tatlmain specimens retain, in addition to a titanomagnetite or multidomain magnetite component, a pseudo-single (PSD) or single domain (SD) magnetite, pyrrhotite or hematite remanence component.

4.3 Saturation remanence tests

SIRM tests revealed distinct differences in the magnetic mineralogy of the specimens from the Ragged and Tatlmain plutons. Six specimens from sites 2, 4, 6, 10, 11 and 13 in the Ragged stock gave low saturation magnetization (J_{sat}) values of 200, 121, 170, 143, 238 and 275 mT, respectively, where J_{sat} is defined as 90 per cent of the magnetization intensity after pulse magnetizing in a direct

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Table 1. Site location, sampling and characteristic remanent magnetization (ChRM) data.

Pluton	Site	Location		Median NRM intensity (mA m^{-1})	Specimens measured, endpoints		Characteristic remanent magnetization (ChRM)				
		Latitude (°N)	Longitude (°W)				Declination (°)	Inclination (°)	Precision parameter k	Confidence α_{95}°	Notes
Ragged	1	62.6580	135.6150	1.76	12	9	50.0	66.6	17.9	12.1	
"	2	62.6580	135.6150	1.43	13	5	66.6	68.2	31.4	13.9	
"	3	62.7003	135.6193	20.6	13	7	288.1	77.1	27.5	11.7	
"	4	62.7003	135.6193	21.3	14	12	325.9	82.4	41.5	6.8	
"	5	62.6828	135.6063	1.04	12	9	6.0	70.8	24.8	10.5	
"	6	62.6828	135.6063	2.23	12	8	19.7	62.3	85.5	6.0	
"	7	62.6503	135.5508	61.0	11	10	6.4	79.1	38.1	8.4	
"	8	62.6503	135.5508	5.04	11	6	6.0	73.7	36.6	11.2	
"	9	62.6397	135.5460	1.17	14	7	2.0	70.7	29.6	11.3	
"	10	62.6397	135.5460	75.9	11	2	304.5	65.2	56.7	33.8	a
"	11	62.6522	135.5930	1.19	12	7	2.4	65.8	24.9	12.3	
"	12	62.6522	135.5930	1.02	13	6	338.7	67.5	56.6	9.0	
"	13	62.6570	135.6050	0.548	10	5	306.5	55.5	56.9	10.2	b
"	14	62.6570	135.6050	0.837	11	9	52.1	73.3	28.8	9.8	
Tatlmain	15	62.7487	135.7467	96.8	14	13	325.0	68.3	47.1	6.1	
"	16	62.7487	135.7467	84.7	11	9	26.8	74.1	25.6	10.4	
"	17	62.7347	135.7208	27.9	11	10	14.3	78.8	33.7	8.4	
"	18	62.7347	135.7208	11.9	13	10	32.4	76.1	65.9	6.0	
"	19	62.5912	135.6798	0.347	12	8	26.9	70.9	18.1	13.4	
"	20	62.5912	135.6798	1.43	12	11	42.0	73.1	23.5	9.6	
"	21	62.6300	135.7587	1.42	13	12	15.5	73.5	57.6	5.8	
"	22	62.6300	135.7587	1.25	11	9	4.7	73.9	35.5	8.8	
"	23	62.8495	136.2237	0.228	12	6	5.1	71.5	27.1	13.1	
"	24	62.8495	136.2237	0.683	10	3	51.2	60.7	21.1	27.5	a
"	25	62.8539	136.1735	0.184	9	3	15.8	67.4	41.8	19.3	a
"	26	62.8539	136.1735	0.381	13	6	30.9	75.2	24.6	13.8	
"	27	62.8803	136.2865	1.57	9	5	333.0	59.1	52.2	10.7	
"	28	62.8803	136.2865	8.77	10	3	341.3	62.2	14.6	33.6	a
"	29	62.8542	136.2265	0.233	12	5	342.4	56.5	29.2	14.4	
"	30	62.8542	136.2265	0.278	14	6	355.2	71.2	41.4	10.5	
"	31	62.7923	136.0212	1.07	10	9	344.1	73.6	46.4	7.6	
"	32	62.7923	136.0212	1.96	13	12	17.2	74.6	40.3	6.9	
"	33	62.7217	135.8145	0.73	10	7	356.7	66.5	29.9	11.2	
"	34	62.7217	135.8145	1.45	10	7	19.2	74.1	88.8	6.4	
Ragged, Mean, 12 sites, 62.6634°N, 135.6058°W							13.1	74.3	45.0	6.5	
Tatlmain, Mean, 17 sites, 62.7657°N, 135.9652°W							3.8	72.4	76.5	4.1	

Notes: Site mean ChRM directions excluded from collection mean directions because (a) they represent <5 specimen vectors or (b) they yield an aberrant direction from the main vector population.

field of 900 mT (J_{900}), that is, $J_{sat} = 0.9J_{900}$ (Fig. 6a). The six J_{sat} values have a median J_{sat} at a field of $M = 185$ mT ($Q_1 = 143$ mT, $Q_3 = 238$ mT). The rapid J_{sat} acquisition by the six specimens in a field of ≤300 mT shows that the remanence is carried dominantly by PSD to SD magnetite. The very slow increase in J/J_{900} from ∼100 to 900 mT (Fig. 6a) and the residual J/J_{900} intensity at 150 mT (Fig. 6b) for four of the specimens indicates the presence of a very minor percentage of hematite in the specimens. Further, when the SIRM acquisition curves are combined with the subsequent AF demagnetization intensity curves, the crossover points plot in the PSD-SD magnetite domain (Symons & Cioppa 2000; Figs 6b and c).

Specimens from 14 sites in the Tatlmain pluton were subjected to SIRM testing. They yield J_{sat} values ranging from 310 to 810 mT from two-stage acquisition curves that give a rapid initial rise in remanence intensity from J_0 followed by a slower secondary rise to J_{900} (Fig. 7a). The rapid initial rise is attributed to PSD-SD titanomagnetite or magnetite mostly, whereas the slower secondary rise is attributed to hematite mostly. In general, the higher the J_{sat} value, the greater the hematite contribution to the remanence and the greater the residual magnetization after AF demagnetization at 130 mT (Fig. 7b).

4.4 Progressive TH demagnetization

Progressive TH step demagnetization of the Ragged specimens typically shows some combination of two endmember behaviours. One endmember exhibits a progressively increasing rate of NRM intensity decrease to a temperature of ∼580 °C that indicates the likely presence of SD or PSD magnetite (Fig. 8a). The second endmember shows a near-linear decrease of NRM intensity to a temperature of ∼400 °C that suggests the likely presence of titanomagnetite (i.e. $Fe_{2.8}Ti_{0.2}O_4$) with trace amounts of hematite, providing a trail of points close to the axis between ∼400 and ∼580 °C (Fig. 8b;

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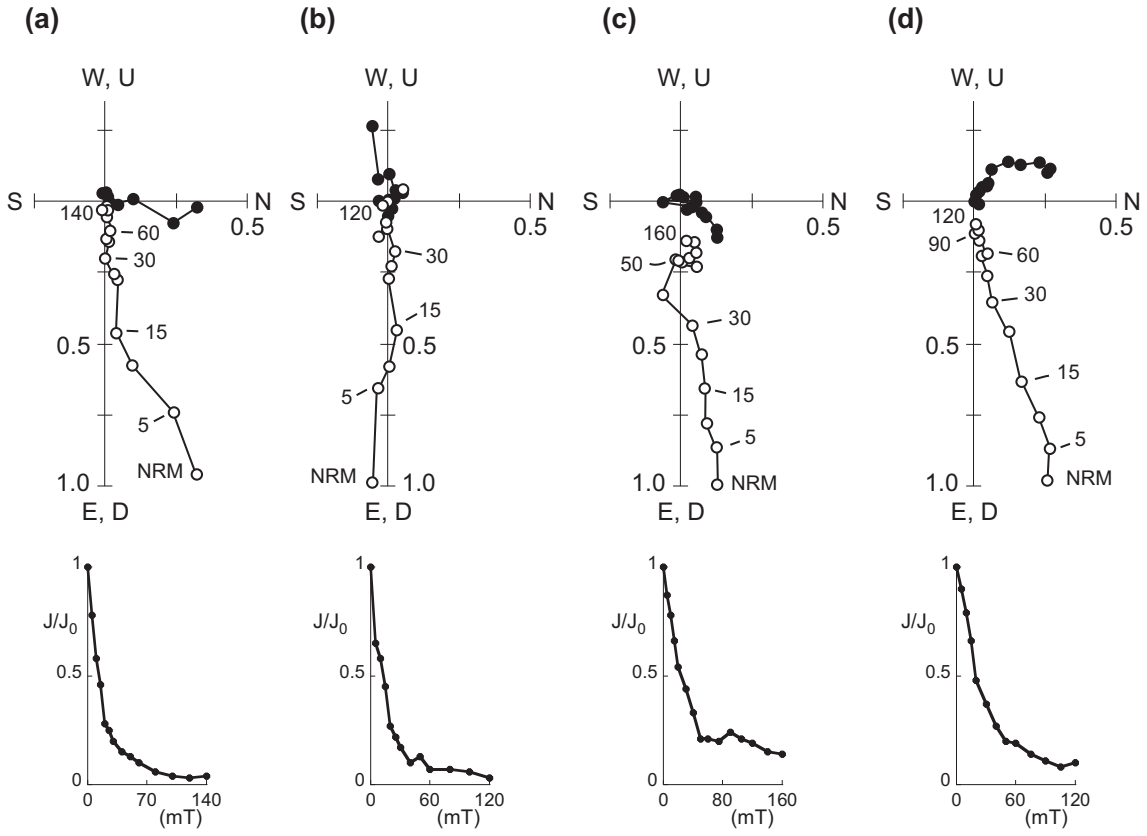


Figure 4. Vector component plots for example specimens from Ragged stock (a and b) and Tatlmain batholith (c and d), showing the isolation of their characteristic remanent magnetization (ChRM) directions on alternating field (AF) step demagnetization. Directions in the horizontal plane (solid circles) lie in the N–E–S–W (north–east–south–west) plane and directions in the vertical plane lie in the N–D–S–U (north-down-south-up) plane. Axial intensities are given as a ratio of the natural remanent magnetization (NRM) intensity ($J_0 = 1.0$), which are (a) 8.28×10^{-2} A m^{-1}, (b) 1.82×10^{-2} A m^{-1}, (c) 1.13×10^{-3} A m^{-1} and (d) 9.98×10^{-4} A m^{-1}. The AF intensity (H) and some labelled steps are in milliteslas (mT).

Özdemir & O'Reilly 1981). Most specimens exhibit a combination of the two endmember behaviours with the addition of (1) erratic changes in remanence vector direction and intensity between ~400 and ~580 °C when the J/J_{900} ratio drops below ~0.15 and (2) a small residual hematite component remaining above ~580 °C at 605 ± 5 °C with a J/J_{900} ratio of <0.05 (Fig. 8c). A very few Ragged specimens exhibit a pronounced kink or J/J_{900} intensity drop in the 250–325 °C range that suggests the presence of trace amounts of pyrrhotite (Fig. 8a).

The Tatlmain specimens display somewhat similar characteristics to the Ragged specimens on progressive TH step demagnetization. Fig. 8(d) exhibits one endmember pattern for the Tatlmain specimens in which the initial decay down to ~0.40 J/J_{900} implies the presence of SD or PSD titanomagnetite with a projected Curie point temperature of ~450 °C. The slow linear decay from ~400 °C to ~580 °C records the progressive linear demagnetization of both magnetite and hematite with hematite being responsible for the ~20 per cent of the remanence remaining undemagnetized at 610 °C. It is important to note that the magnetite and hematite carry virtually identical ChRM directions. The second endmember behaviour found in the Tatlmain specimens displays a rapid near-linear decay from 0 to ~300 °C, likely as titanomagnetite loses its remanence, and then displays a slow near-linear decay from ~300 to 610 °C as magnetite progressively loses its remanence (Fig. 8e). As previously, both the titanomagnetite and magnetite yield the same NRM direction. Overall, most Tatlmain specimens exhibit some combination of both endmember behaviours that likely reflects varying percentages of Ti in the magnetite and of magnetite versus hematite (Fig. 8f). Again note the consistent ChRM direction throughout the entire NRM decay process.

4.5 Mean ChRM directions

The specimen ChRM directions formed a single significant cluster for each site from which the site mean ChRM direction was calculated (Table 1). Four site means (sites 10, 24, 25 and 28) with fewer than five contributing specimen vectors were deemed to be insufficiently represented and rejected from further statistical analysis. Also, one site mean (site 13) gave an aberrant direction of more than six angular standard deviations relative to the mean of the remaining 12 Ragged stock site mean directions and so site 13 was also rejected from further statistical analysis. The stock's remaining 12 site mean directions give a unit mean ChRM direction of declination (D) of 13.1°, inclination (I) of 74.3° (number $N = 12$, resultant $R = 11.8$, precision parameter of Fisher (1953) $k = 45.0$, radius of cone of 95 per cent confidence $\alpha_{95} = 6.5°$; Fig. 9a), and the remaining 17 site mean ChRM directions from the Tatlmain batholith give a unit mean ChRM direction of $D = 3.8°$, $I = 72.4°$ ($N = 17$, $R = 16.8$, $k = 76.5$, $\alpha_{95} = 4.1°$; Fig. 9b). Before tectonic correction, the Ragged stock gives a palaeopole location at latitude

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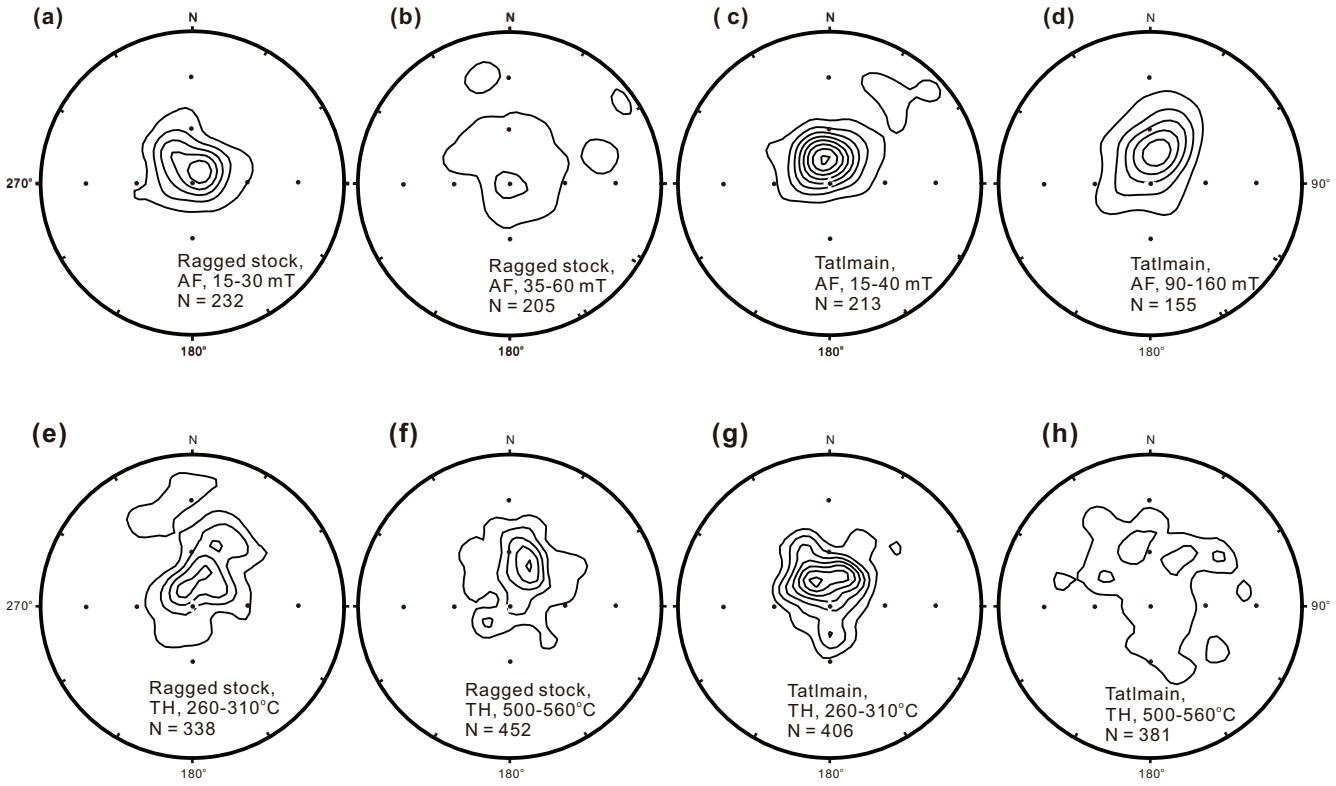


Figure 5. Kamb (1959) equal-area stereocontour plots for all measured specimen remanence vectors (N) upon progressive alternating field (AF) or thermal (TH) demagnetization in the shown step range of milliteslas (mT) or degrees Celsius (°C) from the Ragged stock (a, b, e, f) and Tatlmain batholith (c, d, g, h).

83.5°N, longitude 326.4°E with a radius of 95 per cent confidence for the unit's pole position of $A_{95} = 11.3°$, and the Tatlamin batholith gives a palaeopole at 84.5°N, 22.3°E ($A_{95} = 6.9°$).

5 DISCUSSION

5.1 Magnetic remanence

Progressive AF and TH demagnetization showed that the ChRM of both Ragged stock and Tatlmain batholith was carried by PSD-SD magnetite and titanomagnetite with a composition of about $Fe_{2.8}Ti_{0.2}O_4$ (Özdemir & O'Reilly 1981). In addition, Ragged stock carries a trace component of hematite and Tatlmain batholith carries a trace to ∼25 per cent ChRM component of hematite. Further, the magnetite, titanomagnetite and hematite give near-collinear ChRM directions in a given specimen, indicating that all three minerals were crystallized or recrystallized at the same time.

Magnetite (Fe_3O_4) and titanomagnetite ($Fe_{z-x}Ti_xO_4$) with $x \approx 0.6$ form a solid solution series. Both minerals are common in volcanic rocks, especially submarine basalts, and give Curie temperatures of 580 and 150 °C, respectively (Dunlop & Özdemir 1997). Metamorphic heating in an oxidizing environment causes the titanomagnetite to invert irreversibly to titanomaghemite and ultimately to exsolve to a mixture of magnetite, titanomagnetite, ilmenite, hematite, titanohematite and pseudo-brookite. With sufficient heating, any residual primary magnetite and resulting magnetite, titanomagnetite, hematite and titanohematite acquire a secondary remagnetization thermochemical remanent magnetization (TCRM) with a common ChRM direction. Thus, we interpret both Ragged stock and Tatlmain batholith to retain a secondary TCRM from metamorphism. The finding of a dominant magnetite—titanomagnetite—hematite remanence association has not been previously reported in YTT palaeomagnetic studies, however, it is not surprising given that the lower Finlayson Assemblage is deemed to be a volcanic arc tectonic assemblage (Colpron *et al.* 2006b).

When the unit mean ChRM directions for Ragged and Tatlmain plutons are compared using the correlation test of McFadden & Lowes (1981), the calculated statistic $F = 0.02$ is much less than the 95 per cent confidence comparison statistic of $F_{2,54} = 3.17$, which indicates that their ChRM's are not significantly different at 95 per cent confidence and were likely acquired in the same metamorphic event.

5.2 Tectonic considerations

5.2.1 Study area translation correction

It is necessary to make several tectonic corrections before comparing palaeopoles from the Tatlmain and Ragged plutons to the North American APWP of Torsvik *et al.* (2012) to test for translation or rotation of the YTT relative to the North American craton. First, our study area in the YTT was translated NW by ∼490 km relative to the craton by dextral displacement on the Tintina fault zone between ∼100 and ∼50 Ma (Gabrielse *et al.* 2006; McCausland *et al.* 2006; Symons *et al.* 2017). This correction moves the mean location of the study area from 62.7°N, 224.2°E to 59.6°N, 231.0°E relative to North America in the present day co-ordinates.

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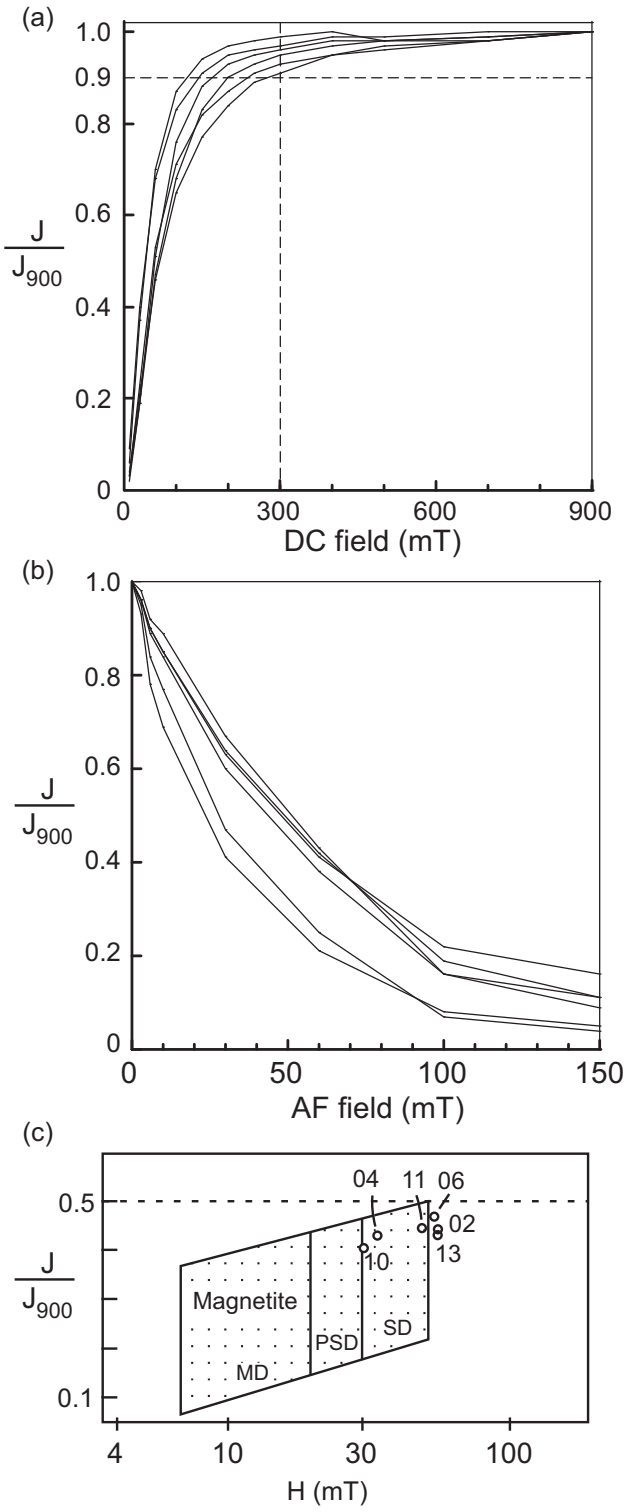


Figure 6. Examples of saturation isothermal remanent magnetization (SIRM) intensity for six Ragged stock specimens showing (a) their SIRM acquisition curves in a direct current (DC) field, (b) their subsequent alternating field (AF) demagnetization curves and (c) the crossover points of the acquisition and demagnetization curves on a 'crossover' plot (Symons & Cioppa 2000). J/J_{900} is the measured intensity (J) to the saturation remanence at 900 mT (J_{900}). H is the applied magnetic field in milliteslas (mT). SD, PSD and MD denote the expected response fields for single, pseudo-single and multidomain magnetite, respectively. The response indicates that Ragged stock's ChRM is carried mostly by SD magnetite or low-Ti magnetite.

5.2.2 *Batholith rotation corrections*

The Tatlmain batholith has been deformed into a sigmoidal shape with its northwestern and central limbs giving different ChRM directions (Fig. 2). The northwestern limb trends WNW (∼305°) in declination and its seven sites (sites 23, 26, 27, 29–32) give a mean ChRM direction of $D = 353.6°$, $I = 69.8°$ ($N = 7$, $k = 60.4$, $\alpha_{95} = 7.8°$), and the central limb trends NNW (∼345°) and its 10 sites (sites 15–22, 33, 34) give a significantly different mean ChRM direction of $D = 12.7°$, $I = 74.1°$ ($N = 10$, $k = 116.3$, $\alpha_{95} = 4.5°$) at 95 per cent confidence (McFadden & Lowes 1981). The sigmoid's vestigial southeastern limb is truncated by the Big Salmon fault zone (Fig. 2). It is likely that the batholith's sigmoidal shape is a response to the two stages of deformation observed in the lower Finlayson Assemblage host rocks (M. Colpron, personal communication, 2018). Palaeomagnetic methods have long been used to demonstrate tectonic rotation in apparently massive plutons (Symons 1977). Further, a palaeomagnetic fold test can be used to determine if the ChRM in a deformed rock unit is pre-folding, syn-folding or post-folding (Graham 1949; McFadden & Jones 1981; Butler 1992). Thus, tectonic corrections are necessary for the folding to remove the effects of rotation in the Tatlmain batholith.

The northwestern limb of Tatlmain batholith parallels the regional tectonic trend of folds and faults in the YTT and surrounding terranes of central Yukon. The change in tectonic trend is ∼40° clockwise (cw) on going from the northwestern limb's ∼305° regional trend to the central limb's ∼345° anomalous trend. Running a declination-only fold test by rotating Tatlmain's central limb counter-clockwise by 19° ± 4° of the 40° trend change brings the two mean ChRM directions into best alignment. The fold test result implies that the ChRM is synfolding and was acquired when rotational deformation was about 50 ± 10 per cent complete. The result also implies that the ChRM inclination of the central limb was further steepened by 4.3° during the original rotation. Given that Tatlmain's ChRM is mostly retained in titanomagnetite at unblocking temperatures of ≤500 °C, it is evident that the syndeformational ChRM is probably recording the regional Early Jurassic metamorphic cooling and deformational episode on average. This timing suggests that the anomalous central-limb trend was caused by the bending of the batholith and its host rocks around the northwest end of the southeastern slice of pericratonic Cassiar terrane as the IMB impacted on southwest side of the YTT (Fig. 2). The fold test results suggest that the likeliest estimate for Tatlmain's ChRM direction after rotation correction but before tilt correction is given by (1) accepting the measured ChRM directions for sites on the regionally trending northwestern limb, (2) fully correcting the measured ChRM directions of sites on the anomalously trending central limb by subtracting 40° from the declination and 9° from the inclination and (3) combining the resulting 17 site mean ChRM directions into one unit mean direction. The result of this two-step correction process gives a unit mean direction before tilt correction for the Tatlmain batholith of $D = 340.9°$, $I = 67.6°$ ($N = 17$, $k = 54.8$, $\alpha_{95} = 4.9°$). Further, Ragged stock is located beside Tatlmain batholith's central limb (Fig. 2). The stock's mean ChRM direction of $D = 13.1°$, $I = 74.3°$ ($N = 12$, $k = 45.0$, $\alpha_{95} = 6.5°$) is virtually identical to the batholith's central limb direction of $D = 12.7°$, $I = 74.1°$ ($N = 10$, $k = 116.3$, $\alpha_{95} = 4.5°$) before rotation correction. Therefore, the same correction procedure is used for the Ragged stock sites as used for the Tatlmain's central limb sites, giving a rotation-corrected unit mean ChRM direction for Ragged stock of $D = 333.1°$, $I = 65.3°$ ($N = 12$, $k = 45.0$, $\alpha_{95} = 6.5°$) before tilt correction.

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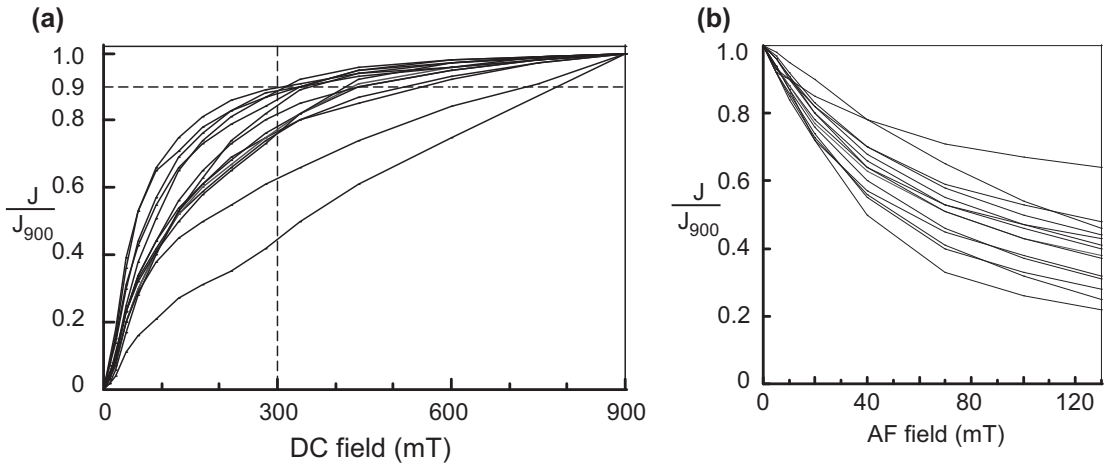


Figure 7. Examples of saturation isothermal remanent magnetization intensity for 14 Tatlmain batholith specimens showing (a) their SIRM acquisition curves in a DC field and (b) their subsequent AF demagnetization curves. Plotting conventions as in Fig. 6.

5.2.3 Tilt corrections

Finally, it is necessary to correct both the Tatlmain and Ragged ChRM directions for tectonic tilts that postdate the acquisition of the ChRM. The ∼70 Ma Carmacks volcanics unconformably overlie the Tatlmain batholith and dip at 5°NE (Tempelman-Kluit 1984). This tilt postdates the formation of Tatlmain's and Ragged's TCRMs but predates the post-70 Ma tectonic translation of the YTT. Thus 5°NE is the initial required tilt correction. Further, the metasediments of the host lower Finlayson Assemblage outcrop along the northeastern side of the Tatlmain batholith. These metasediments dip at 20°–40° NE (M. Colpron, personal communication, 2018), or at 30° ± 7° NE assuming a normal distribution of the dip values. The 30° ± 7° NE value includes the 5° NE post-70 Ma tilt, leaving 25° ± 7° NE for pre-70 Ma tilt as a maximum value. Guided by the results of the 'declination-only' fold test that shows that Tatlmain batholith's synfolding ChRM was acquired when folding was ∼50 per cent completed of the 25° ± 7°, the best estimate for tilting at the time of ChRM acquisition is 13° ± 7° NE. Combining the 5° NE and 13° ± 7° NE corrections gives a 'best estimate' for tilt correction of 18° ± 7° NE. Using the 305° regional structural trend for strike and the 18° tilt correction gives an estimated tilt-corrected ChRM direction of $D = 3.2°$, $I = 54.1°$ ($\alpha_{95} = 4.9°$) for Tatlmain batholith and of $D = 356.1°$, $I = 53.4°$ ($\alpha_{95} = 6.5°$) for Ragged stock. The combination of a 305°-trending northwestern limb dipping at 18°N and a 345°-trending central limb dipping at 18°E provides an estimate that the Tatlmain batholith has been deformed into an anticlinal shape that plunges at ∼22° towards N55°E. Finally, using a mean site location of 231.0°E, 59.6°N to correct for post-100 Ma translation on the Tintina fault zone and the tilt-corrected unit mean ChRM directions, the tectonically corrected palaeopoles are located at 64.9°N, 44.8°E ($A_{95} = 5.9°$) for the Tatlmain batholith and at 64.2°N, 58.5°E ($A_{95} = 7.7°$) for the Ragged stock.

5.2.4 Fit of the Tatlmain and Ragged palaeopoles to North America

Given the imprecision of the structural attitudes and the complexity of the tectonic correction process, it is evident that the Tatlmain and Ragged palaeopoles may not be deemed to be suitable for use in the construction of a conventional continental APWP. We argue, however, that the two palaeopoles are sufficiently reliable to support a number of conclusions. First, the Tatlmain and Ragged palaeopoles are notably closer to the 190 and 200 Ma palaeopoles on the North American APWP of Torsvik *et al*. (2012) than to any of the path's other palaeopoles (Fig. 10). This closeness supports the conclusion that Tatlmain batholith and Ragged stock acquired their TCRMs during the Early Jurassic metamorphic and orogenic event. Thus, the palaeopoles' ages agree with the 196 ± 6 Ma average of nine Ar/Ar and one K-Ar radiometric determinations from muscovite and biotite samples from the Tatlmain batholith and Finlayson Assemblage reported for the study area (Breitsprecher *et al.* 2004; Colpron *et al.* 2007b). Also the closeness supports the conclusions that the batholith's sinusoidal outcrop pattern is structural in origin, and that the stock was rotated tectonically after ChRM acquisition but during post-emplacement deformation. Second, based on their ChRM inclinations of 54.1° and 53.4°, the Tatlmain and Ragged plutons were located at palaeolatitudes of 35° ± 5° and 34° ± 7°, respectively. Relative to the study area, the 200 and 190 Ma palaeopoles on the North American APWP were similarly located at palaeolatitudes of 36° ± 3° and 38° ± 4°, respectively. This co-incidence of palaeolatitudes, is independent of the accuracy of the fold-limb declination estimates, and strongly supports the conclusion that the YTT was located close to the translation-corrected collecting site of 59.6°N, 231.0°E along North America's cratonic margin at 196 ± 6 Ma.

Relative to the 196 Ma reference palaeopole for North America of Torsvik *et al.* (2012), the Tatlmain and Ragged plutons have undergone NW translations of 3.1° ± 5.3° and 3.7° ± 6.6°, using the calculation method of Demarest (1983), as well as cw rotations of 18.7° ± 6.5° and 11.6° ± 8.1°, respectively. The 3.1° ± 5.3° and 3.7° ± 6.6° estimates support the conclusion that the YTT was close to the pluton's pre-70 Ma translation-corrected collecting site on North America's cratonic margin or perhaps ∼200 km further southeast at 196 Ma. Also, the 18.7° ± 6.5° and 11.6° ± 8.1° cw rotation estimates indicate that both the Tatlmain batholith and Ragged stock underwent about 15° ± 7° more cw rotation than accounted for by the rotation corrections. This rotation undercorrection is not surprising given the lack of marker horizons in the two plutons.

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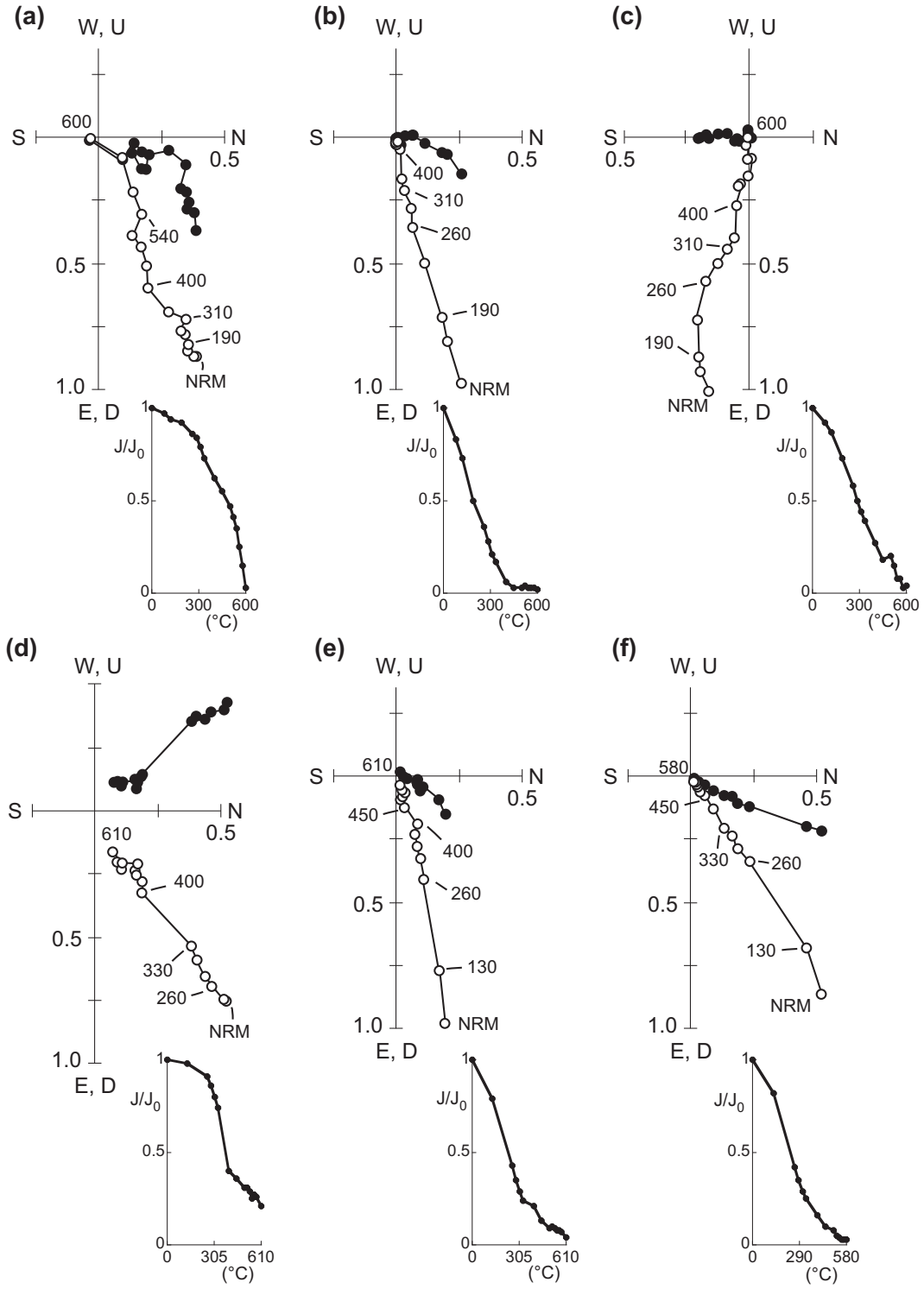


Figure 8. Vector component plots for example specimens from Ragged stock (a–c) and Tatlmain batholith (d–f) showing the isolation of their ChRM directions on thermal (TH) step demagnetization. Plotting conventions as in Fig. 4. The J_{NRM} values are (a) 1.18×10^{-3} A m^{-1}, (b) 6.99×10^{-2} A m^{-1}, (c) 1.76×10^{-2} A m^{-1}, (d) 2.70×10^{-3} A m^{-1}, (e) 1.68×10^{-3} A m^{-1} and (f) 3.75×10^{-3} A m^{-1}.

5.3 Yukon-Tanana terrane's palaeowander path

There are 13 palaeopoles from the YTT for the Late Triassic to present (Table 3). More recent radiometric age dating information is now available for some of the older palaeopoles. The Big Creek batholith has yielded a ^{206}Pb/^{238}U zircon crystallization age of 185.0 ± 0.1 Ma and four K/Ar hornblende ages of 189 ± 4, 184 ± 7, 152 ± 7 and 142 ± 10 Ma that define an ∼47 Ma cooling history centred on about 165 Ma (Stevens *et al.* 1982; Hunt & Roddick 1987; M. Colpron, personal communication, 2018). The cooling history is compatible with six Al-in-hornblende geothermobarometric determinations that yield a ∼16 ± 2 km emplacement

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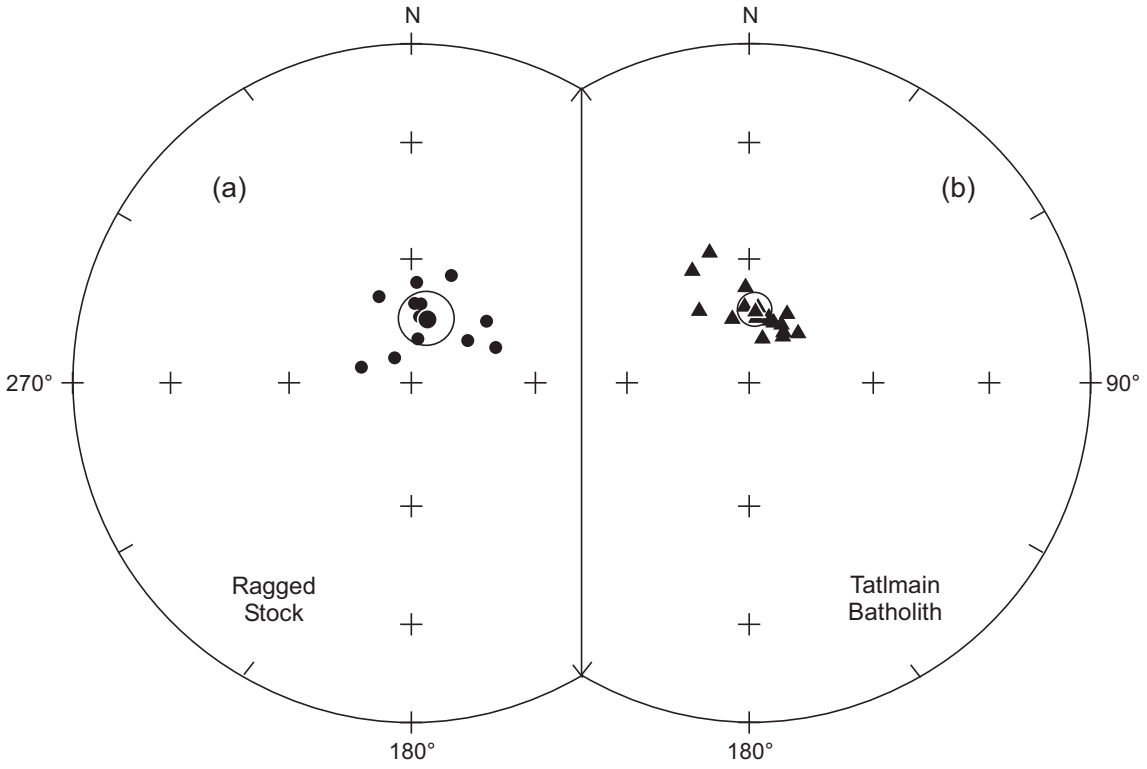


Figure 9. Mean ChRM directions for (a) the 12 accepted sites (●) from Ragged stock and (b) the 17 accepted sites (▲) from Tatlmain batholith as listed in Table 1. The overall unit mean direction is denoted by a larger symbol and surrounded by its cone of 95 per cent confidence.

depth prior to uplift and cooling (Symons *et al.* 2000). Further, the batholith's palaeopole falls on North American's APWP of Torsvik *et al.* (2012) at 165 ± 5 Ma (Fig. 10). Also, Big Creek's palaeopole is significantly different in location from the other four Jurassic poles in Table 2, resulting on a significant increase in the precision parameter (k) from 23.0 to 37.5 on omitting it from the grouping of the other four Early Jurassic palaeopoles. For these reasons, we interpret Big Creek batholith's ChRM to have been acquired during uplift at 165 ± 5 Ma (Parsons *et al.* 2019). Similarly, Granite Mountain batholith has yielded $^{206}Pb/^{238}U$ zircon ages of 194.7 ± 0.1, 197.1 ± 1.2 and 198.5 ± 3.5 Ma with a mean of 196.8 ± 1.3 Ma (M. Colpron, personal communication, 2018).

The only Cenozoic palaeopole for the YTT comes from the 56 ± 1 Ma Pattison pluton that is reported in detail in the Ph.D. thesis of P.J.A. McCausland (2002). Using the calculation method of Demarest (1983), the palaeopole indicates that the extreme southwest side of the YTT was translated 9° ± 7° NW and rotated 28° ± 18° cw after emplacement at 56 Ma relative to the North American craton (Table 3). The remaining 12 Mesozoic palaeopoles form four age groups of 69 ± 1, 101 ± 10, 165 and 198 ± 4 Ma that give mean translation distances of −1° ± 15°, 0° ± 14°, 2° ± 10° and −2° ± 13°, respectively, relative to their respective North America reference palaeopoles (Table 3). None of these four age groups from the Cretaceous and Jurassic are significantly different from the North American reference APWP at > 95 per cent confidence. Overall, the five YTT palaeopole age groups give an average translation distance of 1.6° ± 3.4° NW or 180 ± 380 km. This 180 ± 380 km translation distance agrees with the geological estimate of 490 ± 30 km of Gabrielse *et al.* (2006) for NW displacement of the YTT on the Tintina fault zone during mid-Cretaceous to Eocene time. Further, because the four groups of Mesozoic palaeopoles fit the North American APWP so closely, they give errors for the terrane rotation values that are highly variable or even incalculable. Nonetheless, the 3° ± 4° cw average rotation for the means of the four groups does imply that the YTT overall in Yukon was not rotated significantly with respect to North America since the Late Triassic. The fact that the 12 Cretaceous and Jurassic palaeopoles from the YTT yield both non-significant terrane translation and rotation on average provides powerful evidence that the YTT was accreted solidly to North America by the start of the Jurassic, and that it is para-autochthonous only because of Cretaceous–Eocene translation on the Tintina fault zone. The para-autochthonous conclusion further refines the YTT tectonic model given by Symons *et al.* (2015, 2017) for the Late and mid-Cretaceous and extends the model back in time to the latest Triassic.

5.4 The YTT-IMB tectonic relationship

It has long been evident that terranes along the southern coast of Alaska and the western coast of British Columbia, termed 'Baja Alaska' and 'Baja BC', respectively, have been rafted northwards from mid-palaeolatitudes (Stone 1977; Irving 1985). In their summary of the post-Palaeozoic palaeomagnetic results from the YTT and IMB terranes, Symons *et al.* (2005) noted that the two YTT Jurassic palaeopoles then available (BC and GM in Table 2) indicated that the YTT had 'been close to its present position and unrotated with respect to the craton at least since the Early Jurassic'. Palaeomagnetic results from the Taylor Mountain batholith and now from this study have confirmed that the YTT has been a para-autochthonous terrane of North America since the Late Triassic (Symons *et al.* 2009, 2015), and that is was not part of the hypothesized far-travelled allochthonous 'Baja BC' microcontinent (Enkin

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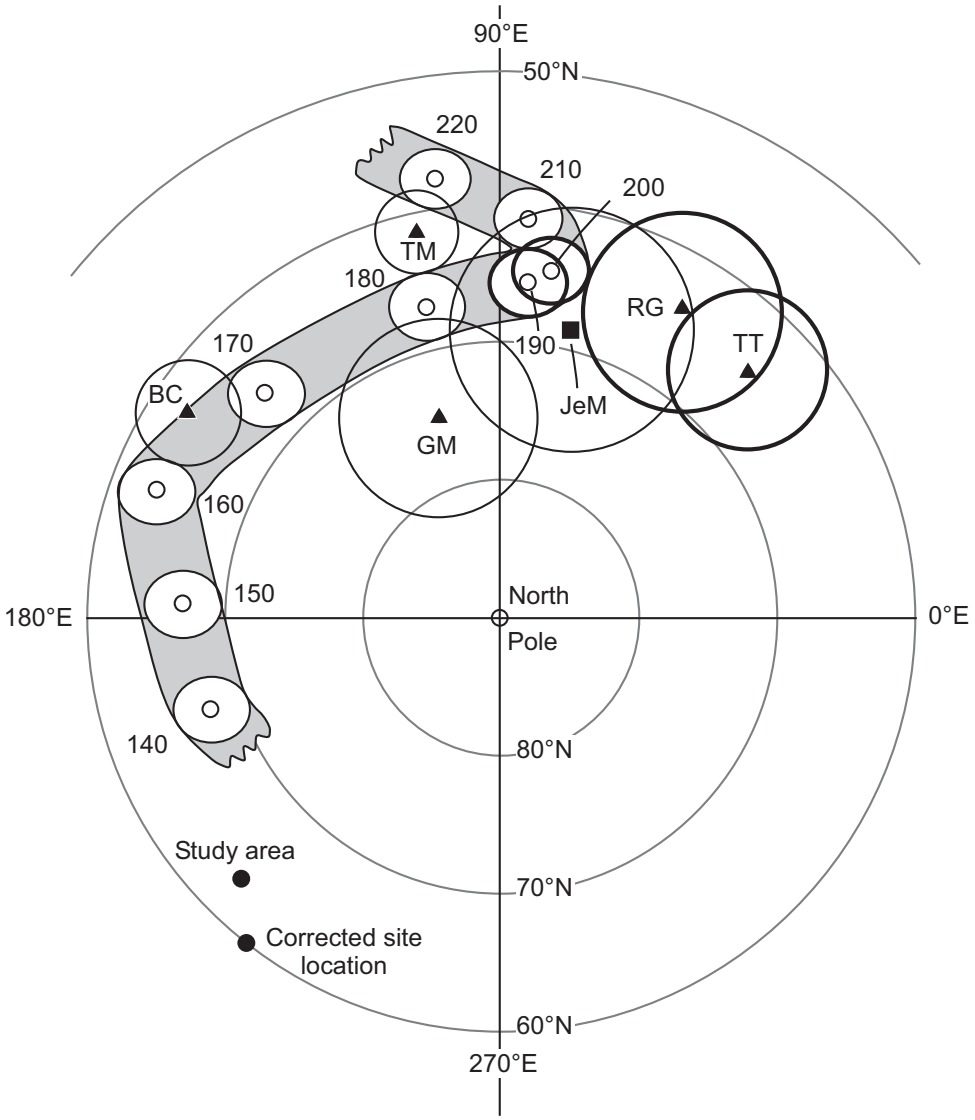


Figure 10. North polar projection of the North America polar wander path of Torsvik *et al.* (2012) from 140 to 220 Ma. Plotted on the path are the tectonically corrected Jurassic palaeopoles from the YTT (Table 2; BC, Big Creek batholith; GM, Granite Mountain batholith; RG, Ragged stock; TM, Taylor Mountain; TT, Tatlmain batholith). J_eM is the location of the Early Jurassic mean palaeopole. The reference palaeopoles on the path are shown with a radius of 3° of arc that match the actual radii for the 190 and 200 Ma palaeopoles. The RG and TT palaeopoles are shown with 2σ and the BC, GM, JeM and TT palaeopoles with 1σ confidence limits.

2006; Enkin *et al.* 2006) or the 'Cordilleran (SAYBIA) ribbon' continent (Johnston 2001; Johnston 2008). However, it still could be that the IMB terranes are far-travelled and allochthonous to both the YTT and the North American craton. It is germane, therefore, to now examine the tectonic relationship between the YTT and IMB terranes.

Symons *et al.* (2005) found that the 15 available IMB palaeopoles from igneous rock units with ages between 3 and 54 Ma showed (1) a non-significant 2° ± 4° (2σ) northward translation and (2) progressive rotation of 0.29° ± 0.11° cw Ma^{-1} or 16° ± 5° cw by 54 Ma relative to North America, which agree closely with the 3° ± 7° N translation and 21° ± 12° cw rotation values for the IMB given in Table 3 at 52 ± 2 Ma using the updated APWP of Torsvik *et al.* (2012).

Symons & Kawasaki (2011) compared the available palaeopoles from the IMB (6 poles), YTT (1 pole) and eastern CPC (11 poles) to the North American APWP. They suggested that the IMB, YTT and CPC were (1) assembled along the western margin of the North American craton by ∼58 Ma as part of 'Baja BC' and (2) rotated 19.3° ± 10.5° cw thereafter, but mostly between ∼50 and ∼45 Ma as massive Eocene plutons were being emplaced along the western CPC belt. Symons & Kawasaki (2011) suggested further that the IMB was driven up and over the western marginal ramp of North America as a thin skin and rotated ∼20° cw in the process. We now conclude that the YTT was not part of 'Baja BC' but was part of the cratonic marginal ramp since the Early Jurassic.

Eight of the 10 available Cretaceous palaeopoles from the IMB form a concordant cluster of tectonic translation and rotation values (Symons *et al.* 2005). The two anomalous palaeopoles come from deformed IMB strata in the eastern CPC of southern British Columbia (McCausland *et al.* 2006). The eight concordant IMB palaeopoles show on average an increase in northward translation (3° ± 7° → 8° ± 14° → 10° ± 9°) and in cw rotation (21° ± 12° → 53° ± 26° → 58° ± 18°) with increasing age (52 ± 2 Ma → 92 ± 10

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Table 2. Late Triassic to Eocene Palaeopoles for the Yukon-Tanana terrane of Yukon and adjacent Alaska.

Geologic unit		Age (Ma)	Number of sites	Location		Unit palaeopole		A_{95}°	References and notes
				Latitude (°N)	Longitude (°E)	Latitude (°N)	Longitude (°E)		
PP	Pattison Pluton	56 ± 1	13	62.4	222.1	86.6	148.1	8.0	1
SE	Seymour Creek Stock	68.5 ± 0.2	16	62.3	222.8	55.2	202.5	8.4	2
PM	Prospector Mountain Stock	68.7 ± 0.4	17	62.5	222.1	77.1	231.7	8.1	3
SW	Swede Dome Stock	69.8 ± 1.3	17	64.2	219.6	78.6	219.5	5.5	4
CV	Carmacks Volcanics	70 ± 3	41	62.1	223.7	75.1	129.0	6.3	5, 6, 7, a
FK	Fort Knox Stock	93 ± 1	18	65.0	212.6	69.2	185.3	5.9	8, b
DR	Dawson Range Batholith	96 ± 4	11	62.6	221.6	80.3	221.7	7.2	9
QL	Quiet Lake Batholith	115 ± 1	14	61.1	226.4	80.2	173.8	10.2	10
BC	Big Creek Batholith	165 ± 5	18	62.2	222.6	62.9	146.6	6.5	11
GM	Granite Mountain Batholith	197 ± 1	8	62.3	223.0	74.7	106.9	12.0	12
RG	Ragged Stock	196 ± 6	12	62.7	224.4	64.2	58.5	7.7	13, c
TT	Tatlmain Batholith	196 ± 6	17	62.8	224.0	64.9	44.8	5.9	13, c
TM	Taylor Mountain Batholith	212 ± 3	17	64.0	217.8	61.4	102.2	5.2	14

Abbreviations as in Table 1. Reversed polarity directions converted to antipodal normal polarity to compute the mean directions.
References: (1) McCausland *et al*. (2006); (2) McCausland *et al*. (2001); (3) Symons *et al*. (2017); (4) McCausland *et al*. (2005); (5) Johnston *et al*. (1996); (6) Wynne *et al*. (1998); (7) Enkin *et al*. (2006); (8) Symons & McCausland (2006); (9) McCausland *et al*. (2006); (10) Symons *et al*. (2015); (11) Symons *et al*. (2000); (12) McCausland (2002); (13) this study; (14) Symons *et al*. (2009).
Notes: (a) Average of 41 sites at scattered locations. (b) Direction includes 121° clockwise correction for the Alaskan orocline. (c) Multiple corrections for tectonic translation, rotation and tilt are incorporated to determine this palaeopole.

Table 3. Latest Triassic to present Terrane motion estimates.

Unit palaeopoles averaged[a]	Mean age (Ma)	Mean location		Mean palaeopole			North American cratonic palaeopole			Translation polewards (°NW)	Rotation clockwise (°)
		°N	°E	°N	°E	A_{95}°	°N	°E	A_{95}°		
Yukon-Tanana terrane											
PP	56 ± 1	62.4	221.1	86.6	148.1	8.0[c]	74.9	186.5	4.0	9 ± 7	28 ± 18
SE, PM, SW, CV	69 ± 1	62.8	222.1	74.6	198.1	18.6	75.2	194.6	4.8	−1 ± 15	−1 ± N[d]
FK, DR, QL	101 ± 10	63.0	220.5	77.2	191.0	12.4	74.9	185.2	12.3	0 ± 14	9 ± 53
BC	165 ± 5	62.2	222.6	62.9	146.6	6.5[c]	64.3	149.8	11.0	2 ± 10	−1 ± 20
GM, TT, RG, TM	198 ± 14	63.3	227.3	68.6	76.2	15.4	65.5	82.4	2.8	−2 ± 13	4 ± 17
Intermontane Belt terranes											
DF. OL, SP, MV, FC[b]	52 ± 2	52.9	236.5	86.3	228.7	7.0	76.2	184.0	4.1	3 ± 7	21 ± 12
ML*b*, SK SZ, CK	92 ± 410	53.0	235.3	71.2	306.7	15.3	75.5	194.8	7.7	8 ± 14	53 ± 26
SP, MM[b], WH[b], AX	112 ± 8	57.1	231.6	73.4	308.4	10.0	73.9	192.6	4.4	10 ± 9	58 ± 18
FD[b], FJ[b]	169 ± 4	59.7	226.3	56.6	279.0	7.8	65.6	139.6	12.2	−9 ± 12	116 ± 22
GB, CM, NV	205 ± 7	49.9	239.3	53.2	11.2	6.0	62.9	84.0	2.4	4 ± 5	41 ± 6

[a]Unit palaeopole acronyms listed for the YTT in Table 2 and for the IMB in McCausland *et al*. (2006).
[b]Identifies palaeopoles from the IMB in Yukon and northernmost British Columbia.
[c]A_{95} is the unit value and not an averaged value.
[d]*N*, number is incalculable.

Ma → 112 ± 8 Ma) of the IMB relative to North America and to the YTT (Table 3). Further, the eight IMB palaeopoles are widely distributed geographically with three from Yukon and the other five spread from central British Columbia to northern Washington state. Thus, there are now 13 Eocene to Late Triassic palaeopoles that show the YTT to be para-autochthonous and 13 Eocene to Cretaceous palaeopoles that show the IMB to have been transported northwards by ∼1100 km and rotated cw by ∼55° since ∼125 Ma. Note also that all of the IMB palaeopoles come from Stikinia except for the palaeopole from the 125 ± 5 Ma Axelgold intrusions that comes from Quesnellia (AX, Table 3), which suggests that Stikinia and Quesnellia formed a coherent tectonic unit by ∼125 Ma.

The interpretation of the IMB's Jurassic translation and rotation data is hampered by the lack of reliable palaeopoles. The three Late Triassic—Early Jurassic palaeopoles (GB, CM and NV, Table 3) give a 205 ± 7 Ma average of 4° ± 5° N translation and 41° ± 4° cw rotation and come from southern British Columbia, and they indicate that the IMB was not moving significantly relative to North America during the ∼205–112 Ma time interval (Table 3).

There are four palaeopoles from the IMB with ages between ∼205 and ∼112 Ma, but they give disparate tectonic motion values. The 148 ± 9 Ma Francois Lake (or Topley) intrusions in Stikinia have given a controversial palaeopole at 58.4°N, 121.8°E (A_{95} = 10.1°) that Symons (1973, 1983) interpreted to be a representative Jurassic

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palaeopole. Lowe *et al.* (2001) concluded from the study of mostly deformed and altered rocks in and near the Endako molybdenite mine that the Jurassic intrusions had suffered Eocene tilting and remagnetization. Symons *et al.* (2005) discussed and disputed their conclusion. Nonetheless, the palaeopole is close to ∼173 Ma on the North American APWP whereas virtually all other ≤130 Ma palaeopoles from Stikinia are cw rotated by tens of degrees (Harris *et al.* 1997a,b, 1999a). One possible explanation may be that YTT or equivalent rocks are exposed in a window through the Stikine terrane. The second deviant Jurassic palaeopole is from the 177 ± 4 Ma Teslin Crossing pluton, which comes from a small collection of sites with too large an A_{95} value of 16.5° to be reliable (Harris *et al.* 1999b). The third and fourth Jurassic palaeopoles come from the 173 ± 3 Ma dikes and 177 ± 4 Ma host rocks of the Fourth of July batholith (Table 3). The dikes and batholith belong to the Cache Creek terrane of the IMB and they give very different translation and rotation values to those found in either Stikinia or Quesnellia (Harris *et al.* 2003).

5.5 Fit of palaeopoles to tectonic models

Coherent terrane APWPs are available for the Canadian Cordillera for only the YTT back through the Early Jurassic and the IMB back through the Early Cretaceous. Any acceptable tectonic model for the Cordillera should give terrane palaeopole locations that align with these APWPs. Here we evaluate the Klondike Orogeny, Intra-Oceanic Arc, Cordilleran Ribbon and Orogenic Wedge models.

The five Jurassic YTT palaeopoles acquired their remanence in three fashions. The Taylor Mountain and Granite Mountain plutons (palaeopoles TM and GM) retain dominantly primary Early Jurassic TRMs upon intrusion. The Big Creek batholith (BC) retains a dominantly secondary Middle Jurassic remagnetization TRM or TCRM of its primary Early Jurassic remanence that was acquired upon uplift. The Ragged stock's and Tatlmain batholith's (RS, TL) primary TRMs were also remagnetized to secondary TCRMs during the Early Jurassic. As discussed earlier in Section 2.5, the Klondike Orogeny model postulates that the YTT collided with the North American plate in the Late Permian with subsequent deposition of an Early-Middle Triassic burial sedimentary assemblage and Late Triassic—Early Jurassic arc sequence that record compression, metamorphism and magmatism (Colpron *et al.* 2006b; Beranek & Mortensen 2011; Nelson *et al.* 2013). Similarly, the Intra-Oceanic Arc model proposes that the YTT collided and accreted with the Slide Mountain terrane in the Middle to Late Permian, that together they collided with the North American plate in the Middle Triassic, and that orogenesis continued through the Early Jurassic (van Staal *et al.* 2018; Parsons *et al.* 2019). Thus both the Klondike Orogeny and Intra-Oceanic Arc models provide the necessary geologic conditions in the Early and Middle Jurassic to yield concordant YTT palaeopoles to the North American APWP, However, neither model describes the subsequent geologic evolution of the YTT in either plate or terrane tectonic terms. We note that the North American APWP exhibits a major cusp in the Early Jurassic but no cusp in the Middle Permian (Torsvik *et al.* 2008, 2012). The APWP cusp is indicative that the North American plate was involved in a major Early Jurassic tectonic collision that caused it to undergo a major change in its drift direction and velocity, which favours the Intra-Oceanic Arc model over the Klondike Orogeny model.

To its credit, the Cordilleran Ribbon model attempts to explain the evolution of the major Mesozoic-Cenozoic tectonic features of the western North American Cordillera (Ross & Ross 1983). Unfortunately, Johnston (2001, 2008) chose to base this model for the Canadian Cordillera solely on the anomalous palaeolatitude of the Carmacks volcanics' palaeopole (Symons *et al.* 2017), and it is anomalous relative to > 25 Mesozoic-Cenozoic palaeopoles from the YTT, IMB and Selwyn basin (Symons *et al.* 2005; Enkin 2006; Symons & Kawasaki 2011; Kawasaki & Symons 2012). Overall, the Cordilleran Ribbon model does not provide a good fit to the APWPs of either the YTT or IMB terrane for any Mesozoic-Cenozoic time period.

To summarize, both the Klondike Orogeny and Intra-Oceanic Arc tectonic models are consistent with the post-Triassic APWP for the YTT by concluding that the YTT accreted to North America prior to the Jurassic. Other than participating in ∼500 km of Cretaceous–Eocene dextral displacement on the Tintina and parallel fault zones, the exposed YTT remained a stable unrotated-overall para-autochthonous terrane of North America and provided the metamorphosed basement for the overlying post-Triassic IMB strata.

Staples et al. (2014, 2016) have proposed an 'Orogenic Wedge' model to explain the Mesozoic tectonic evolution of the northern Canadian Cordillera. They interpret metamorphism, exhumation and deformation in Yukon to be diachronous in both the YTT and IMB strata. They suggest that the YTT and trailing IMB terranes were carried northeasterly (orogen-perpendicular) on thrust faults up and over the subducting western marginal ramp of the North American plate during the Early Jurassic to mid-Cretaceous interval. The five Jurassic YTT palaeopoles, moving only ∼3° along the APWP, are quite consistent with the Orogenic Wedge model (Fig. 10). Staples *et al.* (2016) suggest that the regional tectonic kinematics on the YTT changed in the mid-Cretaceous at ∼110 Ma from orogen-perpendicular compression to orogen-parallel extension. This seems unlikely because plate modelling shows near-orthogonal compression and closure between the Farallon and North American plates of ∼150 ± 50 km Ma^{-1} from about 125 to 90 Ma (Engebretson *et al.* 1985; Kelley 1993). Also, the seven mid-Cretaceous to Palaeocene palaeopoles from the YTT are concordant with the North American APWP after correction for 490 km of dextral Late Cretaceous to Eocene strike-slip translation on the Tintina fault zone (Table 3; Symons *et al*. 2015, 2017).

In contrast to the YTT, the Jurassic palaeopoles from the IMB terranes—Stikinia, Quesnellia and Cache Creek—are widely scattered with no consistent regional pattern of translation or rotation (Symons *et al.* 2005; McCausland *et al.* 2006), suggesting the IMB terranes were fragmented and variably rotated upon initial accretion with each other and/or with the YTT and North America. However, the Cretaceous and younger IMB palaeopoles, mostly from Yukon and British Columbia in Stikinia, do show a consistent pattern from the present back to ∼120 Ma of increasing northward translation up to ∼10° polewards and steadily increasing rotation up to ∼55° cw relative to North America (Table 3). Historically, this pattern of northward translation and cw rotation has been interpreted by palaeomagnetists to signify that the IMB was part of 'Baja BC', having been carried northwards along the western margin of North America during the mid-Cretaceous and then being rotated cw on accretion (Irving & Wynne 1991; Symons *et al.* 2005).

However, if the IMB was accreted to North America by the Early Jurassic as the Klondike Orogeny and Oceanic-Arc models propose, then the 'Baja BC' model is wrong and a new tectonic model is required to explain the IMB palaeopoles. It is also true to point out here that no plausible plate tectonic model has been published yet to explain the Laramide orogeny in the Mackenzie, Ogilvie and Richardson mountain ranges of Yukon. Recently, Hayward

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(2019) has reported a décollement between the upper and lower crust in Yukon and adjacent Alaska and British Columbia. Based on thousands of gravity measurements, the technique distinguishes between lower-density upper crust, composed mainly of felsic plutons and less metamorphosed sediments, and higher-density lower crust composed mainly of more mafic and metamorphosed rocks. In the Selwyn basin the décollement shallows east-southeasterly from a depth of ∼18 ± 3 to ∼11 km beneath the Mackenzie Mountains, and then it is defined by surface faults on the eastern flanks of the Mackenzie, Ogilvie and Richardson mountains. We speculate that the ∼55° cw rotation found in the IMB's Cretaceous palaeopoles is recording upper-crustal Laramide deformation that is not found in the lower-crustal rocks of the YTT. From ∼120 to ∼80 Ma the Yukon region was subject to active Laramide deformation. The Farallon plate was converging perpendicularly against the western margin of North America at a rate of ∼150 km Ma^{-1}, thereby creating northeasterly directed compression with the intrusion of the massive plutons of the outer CPC along the margin (Monger *et al.* 1982). To the north the Arctic Alaskan plate rifted away from the North American plate and rotated by ∼66° ccw to open the Canada basin in the Arctic ocean. Arctic Alaska rotated about a pole located near the Mackenzie delta, potentially causing an increasingly wide wedge of Arctic Alaska to compress against, and overlap with, the western side of Yukon (Fig 11), which could generate south-easterly to easterly compression in Yukon (Carey 1955; Grantz *et al.* 1979; Plafker & Berg 1994). This kinematic scenario could have caused upper-crustal IMB rocks to be rotated cw over lower crustal YTT rocks, and co-incidentally drive the upper crustal rocks easterly to form the Mackenzie, Ogilvie and Richardson mountains. Evidence for Eocene to present cw rotation of the upper crust in Yukon with related shortening and thrust faulting is described in Symons *et al.* (2004) and extended to the mid-Cretaceous in Symons *et al.* (2005). In these papers Symons *et al.* (2004, 2005) also point to evidence for upper-crustal cw rotation and extension in the IMB in southern British Columbia. Note also, if the established ∼500 km of dextral displacement on the Tintina fault extends southwards through British Columbia on the Northern Rocky Mountain Trench fault, then the 'translation poleward' estimates given in Table 3 for the IMB are all decreased by ∼5° to become non-significant, and only the rotation estimates are significant at 95 per cent confidence.

Many papers in Haggart *et al.* (2006) summarize palaeomagnetic, palaeontologic, tectonic, sedimentary and other evidence for hundreds to low thousands of kilometres of northward displacement of the IMB and Insular Belt terranes relative to North America with cw rotations, and subsequent studies continue to add to the concept (Johnston 2008; Hildebrand 2009; Symons & Kawasaki 2011; Szűcs *et al.* 2019)

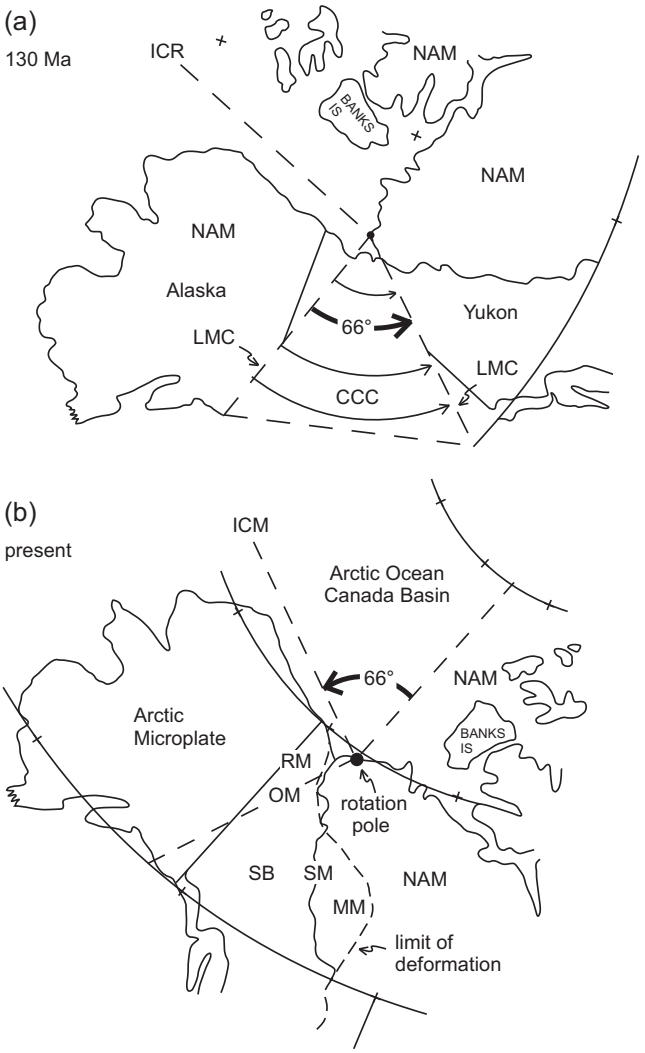


Figure 11. Speculative sketch of the Alaskan microplate rotation during the Laramide deformation. (a) Location of Alaska as part of the North American plate (NAM) prior to ∼130 Ma. An incipient continental rift (ICR) begins to open at ∼130 Ma as Alaska rotates ∼66° ccw away from NAM to open the Canada Basin sphenochasm in the Arctic Ocean (Carey 1955; Plafker & Berg 1994). Most of the ∼66° ccw rotation about a pivot point close to the Mackenzie River delta occurred by ∼90 Ma, but some continues to the present as the Eurasian plate collides with the Alaskan microplate. Assuming a line of maximum compression (LMC) close to the apex of the Alaskan orocline, the ∼66° ccw rotation requires a sphenochasm-sized area of consumed continental crust (CCC) to be destroyed by a combination of (1) obduction through folding, thrust faulting and mountain building; (2) subduction through underplating and melting; and (c) uplift with major erosion and redeposition. In Yukon a combination of '(1)' and '(3)' acting on the 'upper crust' is speculated to be the major contributor to the formation of the Mackenzie (MM), Selwyn (SM), Ogilvie (OM) and Richardson (RM) mountains around the Selwyn Basin (SB).

6 CONCLUSIONS

Palaeomagnetic results from 29 sites have shown that the Mississippian Tatlmain batholith and Ragged stock in the Finlayson Assemblage of the YTT of the Canadian Cordillera carry a very stable TCRM that resides in magnetite, titanomagnetite and hematite. A declination-only palaeomagnetic fold test about a 40° bend in the batholith's outcrop pattern shows that the TCRM is ∼50 per cent synfolding. Multiple $^{40}Ar/^{39}Ar$ and K-Ar radiometric age determinations date the TCRM at 196 ± 6 Ma during a major Early Jurassic deformation and metamorphic event. After tectonic correction for fold rotation, regional dip in the overlying host rocks, and for Late Cretaceous–Eocene translation on the adjacent inboard Tintina fault zone, the TCRM yields palaeopoles of 64.9°N, 44.8°E (A_{95} = 5.9°) and 64.2°N, 58.5°E (A_{95} = 7.7°) for the Tatlmain batholith and Ragged stock, respectively. Both palaeopoles are closest to the 190 Ma and 200 Ma palaeopoles on the North American APWP of Torsvik *et al.* (2012). Further, when grouped by age, 12 of the 13 available Cretaceous and Jurassic palaeopoles for the YTT are not significantly different from their co-eval North

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American palaeopoles. The agreement of the YTT and North American palaeopoles since the latest Triassic asserts that the YTT began accretion to the craton by the Early Jurassic.

Since accretion, the YTT has undergone: (1) Early Jurassic to mid-Cretaceous orogen-perpendicular compression that resulted in palaeomagnetically non-significant ($\leq$ 200 km) northeasterly shortening on thrust sheets and (2) northwesterly Late Cretaceous to Eocene dextral orogen-parallel translation of $\sim$500 km on the Tintina fault zone. In contrast, the Cretaceous and younger strata and plutons of the overlying Stikinia, Quesnellia and Cache Creek (?) terranes of the IMB underwent $\sim$55° cw rotation from the upper-crust rotation of the IMB over lower crustal YTT rocks (Symons *et al*. 2004, 2005) on a recently discovered regional décollement (Hayward 2019) in response to surrounding coeval Laraminde regional tectonic forces associated with (1) the Laramide subduction of the Pacific plate under North America and the related intrusion of the Coast Plutonic Complex; (2) the rotation of the Arctic Alaska plate; (3) the uplift of the arcuate Mackenzie, Ogilvie and Richardson mountain ranges; and (4) subsequent modification by translation along the Tintina and Denali faults.

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REFERENCES

Bailey, R.C. & Halls, H.C., 1984. Estimate of confidence in paleomagnetic directions derived from mixed remagnetization circle and direct observational data, *J. Geophys.,* **54,** 174–182.

Beranek, L.P. & Mortensen, J.K., 2011. The timing and provenance record of the Late Permian Klondike orogeny in northwestern Canada and arc-continent collision along western North America, *Tectonics,* **30**.

Berman, R.G., Ryan, J.J., Gordey, S.P. & Villeneuve, M., 2007. Permian to Cretaceous polymetamorphic evolution of the Stewart River region, Yukon-Tanana terrane, Yukon, Canada: P–T evolution linked with in situ SHRIMP monazite geochronology, *J. Metamorph. Geol.,* **25,** 803–827.

Breitsprecher, K., Mortensen, J.K. & Villeneuve, M.E., 2004. YukonAge 2004: a database of isotopic age determinations for rock units from Yukon Territory, Yukon Geological Survey, CD-ROM.

Butler, R.F., 1992. *Paleomagnetism: Magnetic Domains to Geologic Terranes,* Blackwell Scientific Publications, 319pp.

Calvert, A.J., Hayward, N., Vayavur, R. & Colpron, M., 2017. Seismic and gravity constraints on the crustal architecture of the Intermontane terranes, central Yukon, *Can. J. Earth Sci.,* **54,** 798–811.

Campbell, R.B., 1967. *Geology of Glenlyon Map-area, Yukon Territory (105 L),* Geological Survey of Canada, Memoir 352.

Carey, S.W., 1955. The orocline concept in geotectonics-Part I, *Pap. Proc. R. Soc. Tasmania,* **89,** 255–288.

Colpron, M., 2001. Geochemical characterization of Carboniferous volcanic successions from Yukon–Tanana Terrane, Glenlyon map area (105 L), central Yukon, in *Yukon Exploration and Geology 2000,* pp. 111–136, eds Emond, D.S. & Weston, L.H., Exploration and Geological Services Division, Yukon Region, Indian and Northern Affairs Canada.

Colpron, M., Murphy, D., Nelson, J., Roots, C., Gladwin, K., Gordey, S., Abbott, G. & Lipovsky, P., 2002. Preliminary geological map of Glenlyon (105 L/1-7, 11–14) and northeast Carmacks (115I/9, 16) areas, Yukon Territory (1:125000 scale), Exploration and Geological Services Division, Yukon Region, Indian and Northern Affairs Canada, Open File, 9.

Colpron, M., Gladwin, K., Johnston, S.T., Mortensen, J.K. & Gehrels, G.E., 2005. Geology and juxtaposition history of the Yukon–Tanana, Slide Mountain, and Cassiar terranes in the Glenlyon area of central Yukon, *Can. J. Earth Sci.,* **42,** 1431–1448.

Colpron, M., Mortensen, J.K., Gehrels, G.E., Villeneuve, M. & Nelson, J., 2006a. Basement complex, Carboniferous magmatism and Paleozoic deformation in Yukon-Tanana terrane of central Yukon: field, geochemical and geochronological constraints from Glenlyon map area, in *Paleozoic Evolution and Metallogeny of Pericratonic Terranes at the Ancient Pacific Margin of North America, Canadian and Alaskan Cordillera,* pp. 131–151, eds Colpron, M. & Nelson, J.L., Geological Association of Canada, Special Paper 45.

Colpron, M., Nelson, J. & Murphy, D.C., 2006b. A tectonostratigraphic framework for the pericratonic terranes of the northern Canadian Cordillera, in *Paleozoic Evolution and Metallogeny of Pericratonic Terranes at the Ancient Pacific Margin of North America, Canadian and Alaskan Cordillera,* pp. 1–23, eds Colpron, M. & Nelson, J.L., Geological Association of Canada, Special Paper 45.

Colpron, M., Nelson, J. & Israel, S., 2007a. A transect through the accreted terranes of the northern Canadian Cordillera: from Cassiar, British Columbia to Kluane Lake, Yukon, Arizona Geol. Soc. Field Trip #15, Yukon Geological Survey Open File 2007-3, 84pp.

Colpron, M., Nelson, J. & Murphy, D.C., 2007b. Northern Cordilleran terranes and their interactions through time, *GSA Today,* **17,** 4–10.

Colpron, M., Crowley, J.L., Gehrels, G., Long, D.G., Murphy, D.C., Beranek, L. & Bickerton, L., 2015. Birth of the northern Cordilleran orogen, as recorded by detrital zircons in Jurassic synorogenic strata and regional exhumation in Yukon, *Lithosphere,* **7,** 541–562.

Colpron, M., Israel, S. & Friend, M., 2016a. Yukon plutonic suites, Yukon Geological Survey Open-File 2016–37, scale, 1-750,000.

Colpron, M., Israel, S., Murphy, D., Pigage, L. & Moynihan, D., 2016b. Yukon bedrock geology map, Yukon Geological Survey, Open-File report, 2016-1, scale 1-1,000,000.

Cook, F.A., Clowes, R.M., Snyder, D.B., van der Velden, A.J., Hall, K.W., Erdmer, P. & Evenchick, C.A., 2004. Precambrian crust beneath the Mesozoic northern Canadian Cordillera discovered by Lithoprobe seismic reflection profiling, *Tectonics,* **23**.

Demarest, H.H., 1983. Error analysis for the determination of tectonic rotation from paleomagnetic data, *J. geophys. Res.,* **88,** 4321–4328.

Dunlop, D.J. & Özdemir, Ö., 1997. *Rock Magnetism: Fundamentals and Frontiers,* Cambridge Univ. Press, 571pp.

Engebretson, D.C., Cox, A. & Gordon, R.G., 1985. Relative motions between oceanic and continental plates in the Pacific basin, *Geol. Soc. Am. Spec. Pap.,* **206,** 1–60.

Enkin, R.J., 2006. Paleomagnetism and the case for Baja British Columbia, in *Paleogeography of the North American Cordillera: Evidence For and Against Large-Scale Displacements,* pp. 233–253, eds Haggart, J.W., Enkin, R.J. & Monger, J.W.H., Geological Association of Canada, Special Paper 46.

Enkin, R.J., Johnston, S.T., Larson, K.P. & Baker, J., 2006. Paleomagnetism of the 70 Ma Carmacks Group at Solitary Mountain, Yukon, confirms and extends controversial results: Further evidence for the Baja British Columbia model, in *Paleogeography of the North American Cordillera: Evidence For and Against Large-Scale Displacements,* pp. 221–232, eds Haggart, J.W., Enkin, R.J. & Monger, J.W.H., Geological Association of Canada, Special Paper 46.

Fisher, R.A., 1953. Dispersion on a sphere, *Proc. R. Soc. A*, **217,** 295–305.

Gabrielse, H., 1985. Major dextral transcurrent displacements along the Northern Rocky Mountain Trench and related lineaments in north-central British Columbia, *Bull. geol. Soc. Am.,* **96,** 1–14.

Downloaded from https://academic.oup.com/gji/article/219/3/1660/5570587 by Curtin University Library user on 06 December 2022

Gabrielse, H., Murphy, D.C. & Mortensen, J.K., 2006. Cretaceous and Cenozoic dextral orogen-parallel displacements, magmatism, and paleogeography, north-central Canadian Cordillera, in *Paleogeography of the North American Cordillera: Evidence For and Against Large-Scale Displacements,* pp. 255–276, eds Haggart, J.W., Enkin, R.J. & Monger, J.W.H., Geological Association of Canada, Special Paper 46.

Gradstein, F.M., Ogg, J.G. & Smith, A.G., 2004. *A Geologic Time Scale 2004,* Cambridge Univ. Press.

Graham, J.W., 1949. The stability and significance of magnetism in sedimentary rocks, *J. geophys. Res.,* **54,** 131–167.

Grantz, A., Eittreim, S. & Dinter, D.A., 1979. Geology and tectonic development of the continental margin north of Alaska, in *Developments in Geotectonics,* pp. 263–291, ed. Keen, C.E., Elsevier.

Haggart, J.W., Monger, J. & Enkin, R.J., 2006. *Paleogeography of the North American Cordillera: Evidence for and against Large-scale Displacements,* Cordilleran Section/Geological Association of Canada.

Hansen, V.L., 1990. Yukon-Tanana terrane: A partial acquittal, *Geology,* **18,** 365–369.

Hansen, V.L. & Dusel-Bacon, C., 1998. Structural and kinematic evolution of the Yukon-Tanana upland tectonites, east-central Alaska: a record of late Paleozoic to Mesozoic crustal assembly, *Bull. geol. Soc. Am.,* **110,** 211–230.

Harris, M., Symons, D., Blackburn, W. & Hart, C., 1997a. Paleomagnetic and geobarometric study of the mid-Cretaceous Whitehorse Pluton, Yukon Territory, *Can. J. Earth Sci.,* **34,** 1379–1391.

Harris, M., Symons, D., Blackburn, W. & Hart, C., 1997b. Paleomagnetic study of the mid-Cretaceous Mount McIntyre pluton, Whitehorse map area (105 D), southern Yukon Territory, in *Yukon Exploration and Geology 1996,* pp. 122–130, ed. Roots, C.F., Exploration and Geological Services Division, Yukon, Indian and Northern Affairs Canada.

Harris, M.J., Symons, D.T.A., Blackburn, W.H. & Hart, C.J.R., 1999a. Paleomagnetic and geobarometric study of the Late Cretaceous Mount Lorne stock, southern Yukon Territory, *Can. J. Earth Sci.,* **36,** 905–915.

Harris, M.J., Symons, D.T.A., Hart, C.J.R. & Blackburn, W.H., 1999b. Jurassic plate motions of the Stikine Terrane, southern Yukon: a paleomagnetic and geothermometric study of the Teslin Crossing Pluton (105E/7), in *Yukon Exploration and Geology 1988,* pp. 155–166, eds Roots, C.F. & Emond, D.S., Exploration and Geological Services Division, Indian and Northern Affairs Canada.

Harris, M.J., Symons, D.T., Blackburn, W.H., Hart, C.J. & Villeneuve, M., 2003. Travels of the Cache Creek Terrane: a paleomagnetic, geobarometric and 40Ar/39Ar study of the Jurassic Fourth of July Batholith, Canadian Cordillera, *Tectonophysics,* **362,** 137–159.

Hayward, N., 2019. The 3D geophysical investigation of a Middle Cretaceous to Paleocene Regional Décollement in the Cordillera of Northern Canada and Alaska, *Tectonics,* **38,** 307–334.

Hildebrand, R.S., 2009. *Did Westward Subduction Cause Cretaceous-Tertiary Orogeny in the North American Cordillera?,* Vol. **457,** 71p, Geological Society of America.

Hunt, P.A. & Roddick, J.C., 1987. A compilation of K-Ar dates, Report 17, Geological Survey of Canada, Paper 87-2, pp. 143–210.

Irving, E., 1985. Tectonics: Whence British Columbia?, *Nature,* **314,** 673–674.

Irving, E. & Wynne, P.J., 1991. Paleomagnetic evidence for motions of parts of the Canadian Cordillera, *Tectonophysics,* **187,** 259–275.

Johnston, S.T., 2001. The great Alaskan terrane wreck: reconciliation of paleomagnetic and geological data in the northern Cordillera, *Earth planet. Sci. Lett.,* **193,** 259–272.

Johnston, S.T., 2008. The Cordilleran ribbon continent of North America, *Annu. Rev. Earth Planet. Sci.,* **36,** 495–530.

Johnston, S.T. & Erdmer, P., 1995. Hot-side-up aureole in southwest Yukon and limits on terrane assembly of the northern Canadian Cordillera, *Geology,* **23,** 419–422.

Johnston, S.T., Wynne, P.J., Francis, D., Hart, C.J., Enkin, R.J. & Engebretson, D.C., 1996. Yellowstone in Yukon: the late Cretaceous Carmacks group, *Geology,* **24,** 997–1000.

Kamb, W.B., 1959. Ice petrofabric observations from Blue Glacier, Washington, in relation to theory and experiment, *J. geophys. Res.,* **64,** 1891–1909.

Kawasaki, K. & Symons, D.T.A., 2012. Palaeomagnetism of the Howards Pass Zn-Pb deposits, Yukon, Canada, *Geophys. J. Int.,* **190,** 217–229.

Kelley, K., 1993. Relative motions between North America and oceanic plates of the Pacific basin during the past 130 million years, *M.Sc. thesis*, Western Washington University, Bellingham, Wash.

Kirschvink, J.L., 1980. The least-squares line and plane and the analysis of palaeomagnetic data, *Geophys. J. R. astr. Soc.,* **62,** 699–718.

Lowe, C., Enkin, R.J. & Struik, L.C., 2001. Tertiary extension in the central British Columbia Intermontane Belt: magnetic and paleomagnetic evidence from the Endako region, *Can. J. Earth Sci.,* **38,** 657–678.

McCausland, P.J.A., 2002. Geotectonic motions of the Yukon-Tanana terrane, northern Cordillera: Paleomagnetism and geobarometry of Mesozoic and Cenozoic intrusions, *Ph.D. thesis*, The University of Western Ontario, London, ON, Canada.

McCausland, P.J.A., Symons, D.T.A., Hart, C.J.R. & Blackburn, W.H., 2001. Paleomagnetism and geobarometry of the Granite Mountain batholith, Yukon: minimal geotectonic motion of the Yukon-Tanana Terrane relative to North America, in *Yukon Exploration and Geology 2001,* pp. 163–177, eds Emond, D.S., Weston, L.H. & Lewis, L.L., Yukon Geology Program.

McCausland, P.J.A., Symons, D.T.A. & Hart, C.J.R., 2005. Rethinking "Yellowstone in Yukon" and Baja British Columbia: paleomagnetism of the Late Cretaceous Swede Dome stock, northern Canadian Cordillera, *J. geophys. Res.,* **110,** B12107.

McCausland, P.J.A., Symons, D.T.A., Hart, C.J.R. & Blackburn, W.H., 2006. Assembly of the northern Cordillera: new paleomagnetic evidence for coherent, moderate Jurassic to Eocene motion of the Intermontane Belt and Yukon-Tanana terranes, in *Paleogeography of the North American Cordillera: Evidence for and Against Large-Scale Displacements,* pp. 147–170, eds Haggart, J.W., Enkin, R.J. & Monger, J.W.H., Geological Association of Canada, Special Paper 46.

McElhinny, M.W. & McFadden, P.L., 2000. *Paleomagnetism: Continents and Oceans,* Academic.

McFadden, P.L. & Jones, D.L., 1981. The fold test in palaeomagnetism, *Geophys. J. Int.,* **67,** 53–58.

McFadden, P.L. & Lowes, F.J., 1981. The discrimination of mean directions drawn from Fisher distributions, *Geophys. J. R. astr. Soc.,* **67,** 19–33.

Monger, J.W.H., Price, R.A. & Tempelman-Kluit, D.J., 1982. Tectonic accretion and the origin of the two major metamorphic and plutonic welts in the Canadian Cordillera, *Geology,* **10,** 70–75.

Morneau, Y.E., Gaidies, F., Ryan, J.J. & Zagorevski, A., 2017. Estimates of garnet crystallization and rates of metamorphism for metapelites of the Snowcap Assemblage, Yukon-Tanana Terrane, Yukon, Geological Survey of Canada, Current Research 2017-2, 21, doi:10.4095/304229.

Mortensen, J.K., 1992. Pre-mid-Mesozoic tectonic evolution of the Yukon-Tanana terrane, Yukon and Alaska, *Tectonics,* **11,** 836–853.

Nelson, J.L., Colpron, M., Piercey, S.J., Dusel-Bacon, C., Murphy, D.C. & Roots, C.F., 2006. Paleozoic tectonic and metallogenetic evolution of pericratonic terranes in Yukon, northern British Columbia and eastern Alaska , eds ,Colpron, M. & ,Nelson, J.L., in *Paleozoic Evolution and Metallogeny of Pericratonic Terranes at the Ancient Pacific Margin of North America, Canadian and Alaskan Cordillera,* Geological Association of Canada, pp. 323–360, Special Paper 45.

Nelson, J.L., Colpron, M. & Israel, S., 2013. The Cordillera of British Columbia, Yukon, and Alaska: tectonics and metallogeny , eds ,Colpron, M., ,Bissig, T., ,Rusk, B.G. & ,Thompson, J.F.H.,, in *Tectonics, Metallogeny, and Discovery: The North American Cordillera and Similar Accretionary Settings,* Vol. **17,** pp. 53–109, Society of Economic Geologists.

Oliver, D.H., 1996. Structural, kinematic, and thermochronologic studies of the Teslin suture zone, south-central Yukon Territory, *Unpublished Ph. D. thesis*, Southern Methodist University.

Oliver, D.H. & Mortensen, J.K., 1998. Stratigraphic succession and U-Pb geochronology from the Teslin suture zone, south central Yukon, in *Yukon Exploration and Geology 1997,* pp. 69–75, Yukon Geological Survey Open-File 2016-1, scale 1-1,000,000, Exploration and Geological Services Division, Yukon, Indian and Northern Affairs Canada.

Downloaded from https://academic.oup.com/gji/article/219/3/1660/5570587 by Curtin University Library user on 06 December 2022

Özdemir, Ö. & O'Reilly, W., 1981. High-temperature hysteresis and other magnetic properties of synthetic monodomain titanomagnetites, *Phys. Earth planet. Inter.,* **25,** 406–418.

Parsons, A., Coleman, M., Ryan, J., Zagorevski, A., Joyce, N., Gibson, H. & Larson, K., 2018. Structural evolution of a crustal-scale shear zone through a decreasing temperature regime: the Yukon River shear zone, Yukon-Tanana terrane, Northern Cordillera, *Lithosphere,* **10,** 760–782.

Parsons, A.J., Zagorevski, A., Ryan, J., McClelland, W.C., Van Staal, C.R., Coleman, M.J. & Golding, M.L., 2019. Petrogenesis of the Dunite Peak ophiolite, south-central Yukon, and the distinction between upper-plate and lower-plate settings: a new hypothesis for the late Paleozoic–early Mesozoic tectonic evolution of the Northern Cordillera, *Bull. geol. Soc. Am.,* **131,** 274–298.

Plafker, G. & Berg, H.C., 1994. Overview of the geology and tectonic evolution of Alaska, in *The Geology of North America, G-1. The Geology of Alaska,* pp. 989–1021, eds Plafker, G. & Berg, H.C., Geological Society of America.

Richards, D.R., Butler, R.F. & Harms, T.A., 1993. Paleomagnetism of the late Paleozoic Slide Mountain terrane, northern and central British Columbia, *Can. J. Earth Sci.,* **30,** 1898–1913.

Ross, C.A. & Ross, J.R., 1983. Late Paleozoic accreted terranes of western North America, in *Pre-Jurassic in western North America suspect terranes,* pp. 7–22, ed. Stevens, C.H., Society of Economic Paleontologists and Mineralogists.

Staples, R.D., Murphy, D.C., Gibson, H.D., Colpron, M., Berman, R.G. & Ryan, J.J., 2014. Middle Jurassic to earliest Cretaceous mid-crustal tectono-metamorphism in the northern Canadian Cordillera: recording foreland-directed migration of an orogenic front, *Bulletin,* **126,** 1511–1530.

Staples, R.D., Gibson, H., Colpron, M. & Ryan, J.J., 2016. An orogenic wedge model for diachronous deformation, metamorphism, and exhumation in the hinterland of the northern Canadian Cordillera, *Lithosphere,* **8,** 165–184.

Stevens, R.D., Delabio, R.N. & Lachance, G.R., 1982. Age determinations and geological studies, K–Ar isotopic ages, *Report* **15,** 17–27.

Stone, D.B., 1977. Plate tectonics, paleomagnetism and the tectonic history of the NE Pacific, *Geophys. Surveys,* **3,** 3–37.

Symons, D.T., Harris, M.J., Hart, C.J. & McCausland, P.J., 2006. *Paleomagnetism of the ∼ 91 Ma Deadman Pluton: Post-mid-Cretaceous Tectonic Motion in Central Yukon,* Yukon Exploration and Geology, pp. 299–313.

Symons, D.T.A., 1973. Paleomagnetic results from the Jurassic Topley Intrusions near Endako, British Columbia, *Can. J. Earth Sci.,* **10,** 1099–1108.

Symons, D.T.A., 1977. Geotectonics of Cretaceous and Eocene plutons in British Columbia: a paleomagnetic fold test, *Can. J. Earth Sci.,* **14,** 1246–1262.

Symons, D.T.A., 1983. Further paleomagnetic results from the Jurassic Topley intrusives in the Stikinia subterrane of British Columbia, *Geophys. Res. Lett.,* **10,** 1065–1068.

Symons, D.T.A. & Cioppa, M.T., 2000. Crossover plots: a useful method for plotting SIRM data in paleomagnetism, *Geophys. Res. Lett.,* **27,** 1779–1782.

Symons, D.T.A. & Kawasaki, K., 2011. Eocene rotation of the Coast Plutonic Complex and Intermontane Belt: paleomagnetism of Eocene plutons along the Klondike Highway, *Can. J. Earth Sci.,* **48,** 645–660.

Symons, D.T.A. & McCausland, P.J.A., 2006. Paleomagnetism of the Fort Knox Stock, Alaska, and rotation of the Yukon–Tanana terrane after 92.5 Ma, *Tectonophysics,* **419,** 13–26.

Symons, D.T.A., Williams, P.R., McCausland, P.J.A., Harris, M.J., Hart, C.J.R. & Blackburn, W.H., 2000. Paleomagnetism and geobarometry of the Big Creek Batholith suggests that the Yukon–Tanana Terrane has been a parautochthon since Early Jurassic, *Tectonophysics,* **326,** 57–72.

Symons, D.T.A., Hart, C.J.R., Harris, M.J., McCausland, P.J.A., Sussman, A.J. & Weil, A.B., 2004. Paleomagnetism of the Eocene Flat Creek pluton, Yukon: tectonics of the Intermontane terranes and Mackenzie Mountains, in *Orogenic Curvature: Integrating Paleomagnetic and Structural Analyses,* pp. 187–204, eds Sussman, A.J. & Weil, A.B., Geological Society of America, Special Paper 383.

Symons, D.T.A., Harris, M.J., McCausland, P.J.A., Blackburn, W.H. & Hart, C.J.R., 2005. Mesozoic-Cenozoic paleomagnetism of the Intermontane and Yukon Tanana terranes, Canadian Cordillera, *Can. J. Earth Sci.,* **42,** 1163–1185.

Symons, D.T.A., Kawasaki, K. & McCausland, P.J.A., 2009. The Yukon–Tanana terrane: part of North America at ∼215 Ma from paleomagnetism of the Taylor Mountain batholith, Alaska, *Tectonophysics,* **465,** 60–74.

Symons, D.T.A., McCausland, P.J.A., Kawasaki, K. & Hart, C.J.R., 2015. Post-Triassic para-autochthoneity of the Yukon-Tanana Terrane: Paleomagnetism of the Early Cretaceous Quiet Lake Batholith, *Geophys. J. Int.,* **203,** 312–326.

Symons, D.T.A., Kawasaki, K., McCausland, P.J.A. & Hart, C.J.R., 2017. Paleopole for the 69 Ma Prospector Mountain Stock: A critique of the Carmacks/"Baja BC" transport estimate for Yukon, Canada, *Geophys. J. Int.,* **208,** 349–367.

Szűcs, D., González-León, C.M. & Pálfy, J., 2019. Ammonoid biostratigraphy and paleobiogeography of Lower Jurassic sections from Sonora (northwest Mexico) and their tectonic implications, *J. S. Am. Earth. Sci.,* **93,** 267–282.

Tempelman-Kluit, D.J., 1976. The Yukon crystalline terrane: enigma in the Canadian Cordillera, *Bull. geol. Soc. Am.,* **87,** 1343–1357.

Tempelman-Kluit, D.J., 1984. *Geology, Laberge (105E), Carmacks (115I), Yukon Territory,* Natural Resources Canada.

Torsvik, T.H., Müller, R.D., Van der Voo, R., Steinberger, B. & Gaina, C., 2008. Global plate motion frames: toward a unified model, *Rev. Geophys.,* **46,** 1–44.

Torsvik, T.H. *et al.*, 2012. Phanerozoic polar wander, palaeogeography and dynamics, *Earth-Sci. Rev.,* **114,** 325–368.

van Staal, C.R., Zagorevski, A., McClelland, W.C., Escayola, M.P., Ryan, J.J., Parsons, A.J. & Proenza, J., 2018. Age and setting of Permian Slide Mountain terrane ophiolitic ultramafic-mafic complexes in the Yukon: implications for late Paleozoic-early Mesozoic tectonic models in the northern Canadian Cordillera, *Tectonophysics,* **744,** 458–483.

Wynne, P.J., Enkin, R.J., Baker, J., Johnston, S.T. & Hart, C.J., 1998. The big flush: paleomagnetic signature of a 70 Ma regional hydrothermal event in displaced rocks of the northern Canadian Cordillera, *Can. J. Earth Sci.,* **35,** 657–671.

Zijderveld, J.D.A., 1967. A.C. demagnetization of rocks: analysis of results, in *Methods in Paleomagnetism,* pp. 254–286, eds Collinson, D.W., Creer, K.M. & Runcorn, S.K., Elsevier.

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